

The Mechanics of Growing and Maintaining a Supplier Base: Evidence from Chinese Firms

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Abstract

This paper studies how firms expand and maintain their import supplier base, and how they reconfigure sourcing networks after unexpected disruptions. Using highly disaggregated Chinese transaction data following China's WTO entry, we exploit antidumping (AD) investigations as unanticipated trade shocks and document stylized facts on import sourcing dynamics. We then develop a parsimonious model in which firms face limited search capacity when adjusting sourcing origins. Guided by the model, we estimate the effects of China's AD investigations on firms' entry into and exit from import origin markets. AD shocks induce greater churning for targeted imports: firms simultaneously increase experimentation with new origins and accelerate exit from existing origins. For non-targeted imports, firms become more inert, reducing both entry and exit. These responses are highly heterogeneous: firms with narrower supplier bases and tighter sourcing capacity constraints tend to adjust more aggressively, whereas firms with broader supplier networks exhibit more muted reallocation. Therefore, the trade effects of AD policy extend well beyond the directly investigated products and origins through firms' endogenous adjustment of their supplier bases.

JEL: F13, F14

Keywords: Supplier base, production networks, search capacity, supply chain networks, Antidumping

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1 Introduction

Recent years have underscored the fragility of global supply chains. Firms relying on internationally fragmented production have faced repeated disruptions, including the U.S.-China trade disputes, pandemic-era shutdowns, shipping bottlenecks, geopolitical tensions and sanctions, and abrupt changes in trade policy, that simultaneously raised input costs, delayed deliveries, and forced rapid reallocation across suppliers and locations. These episodes highlight a basic tension at the heart of global sourcing: while international supplier networks can lower costs and expand input availability, they also expose firms to disruptions that are difficult to insure against and costly to manage. Adjusting supplier relationships in response to such shocks requires search, contracting, and coordination, all of which draw on scarce managerial resources and organizational capacity (Park and Ungson, 2001; Rozemeijer, Van Weele and Weggeman, 2003). Despite the policy relevance, we still know relatively little about how firms reoptimize their foreign supplier relationships when shocks hit.

The goal of this paper is to improve our understanding of the constraints that firms face in expanding and maintaining their global supplier base and in responding to unanticipated shocks to this base. We analyze the evolution of Chinese firms' sourcing decisions in the wake of China's entry into the WTO in 2001. With the dramatic expansion of access to global suppliers, firms with little previous experience navigating international markets now had the opportunity to lower their production costs and to upgrade their quality by finding foreign suppliers. As Chinese firms became more inclined to import foreign inputs, many discovered that their newly created relationships with suppliers could be harmed by unanticipated shocks. We use this unique setting to learn how firms manage their supplier base and to obtain insight into the limits of their ability to adjust to shocks quickly.

We first document several facts about the experience of Chinese importers over the period 2002-2015 using transaction-level data from China's customs matched to firm-level survey data. This is a period of dramatic growth in global value chains worldwide in no small part due to China's entry into the WTO. We show that firms accumulate suppliers much as they do tangible capital with important inputs prioritized and depreciated supplier relationships replaced. As firms experience in obtaining foreign suppliers grew, they expended more of their energy replacing existing suppliers than finding suppliers for goods not previously imported. Such dynamic sourcing adjustment is pronounced even at the input level. Each year, the average Chinese importer adds about 21.2% new sourcing origins and subsequently drops roughly 16.7% of existing origins for a given input. Origin entry and exit account for around 20.6% and 14.6% of the input's import value, respectively.

One feature of China’s entry into the WTO that proves useful for understanding the constraints facing firms in sourcing inputs was the burst in antidumping (AD) investigations that followed the sudden reduction in MFN tariffs. The 336 AD incidences (product-origin country-year triplets) that occurred during 2002-2015 can be treated as exogenous from the point of view of the firm and were largely unanticipated because China had little history using AD as a policy prior to its WTO accession. Many of these investigations involved important sources of very specific intermediate inputs for Chinese firms. The sudden introduction of trade barriers hit firms that relied on suppliers named in the AD investigation, while other firms sourced their inputs elsewhere. We find that firms’ efforts to accumulate and maintain suppliers exhibit interdependence across country-product pairs. Firms hit by AD shocks to a particular source for a particular input were quick to find replacements for this supplier but were observed to slow or to suspend altogether their search for new suppliers of other inputs. The pattern of greater origin churning for targeted imports and less churning for non-targeted imports following AD investigations suggests that firms face limited adjustment capacity when responding to supply chain disruptions.

To understand the mechanisms at work in the data, we develop a simple model of a cost-minimizing firm that faces constraints in obtaining access to cheap intermediate inputs. In the model, firms require inputs in multiple industries. By searching for suppliers, a firm can lower its production costs. Once established, a supplier relationship is subject to exogenous shocks that eliminate individual suppliers so that the stock of supplier relationships evolves over time like physical capital in a standard model. The accumulation of new suppliers and the replacement of old suppliers requires firms to allocate resources to search for suppliers. These resources are in fixed supply within the firm, and the stock of internal resources available in any given period is called the firm’s “search capacity.” This stock is an analog for the capacity of managerial offices within the firm that are tasked with procurement.

The model captures the main features of the data. Firms that are just beginning to purchase inputs from abroad begin by prioritizing their most important inputs, i.e. those where foreign inputs are much cheaper and those that account for a larger share of the firm’s input costs. As the firm becomes more embedded in the global supply chain, the set of industries in which it searches expands, with older industries displaying slow net growth but high volumes of supplier replacement. Firms with higher search capacity will ultimately have a higher share of their inputs outsourced and will, everything else equal, have lower marginal costs that would encourage greater domestic and foreign market share.

When hit by a shock to an important existing supplier, such as the antidumping investigations considered in our empirics, firms in our model respond by replacing the

lost suppliers. Intuitively, inputs that the firm already established a supplier for are more important to the firm’s operations than the inputs where a foreign supplier has not already been identified. This means that the firm should be observed to find new suppliers in the affected industry and to limit adjustment elsewhere in order to free up search capacity. This mechanism captures the apparent interdependence across products within the firm that is observed in the data. An important implication of this mechanism is that even though a firm might find a replacement for a shocked supplier, this does not mean that it has fully adjusted to the shock as other input searches were scaled back or put on hold in the meantime.

We formally test the predictions of our model and find strong empirical support for its key mechanisms. First, firms devote more search effort to imports sourced from fewer origins and to imports that account for a larger share of total import expenditure, consistent with an incentive to gradually expand supplier bases. Second, firms with greater search capacity, measured by the number of sourcing origins and the average distance to those origins, source more inputs from abroad. In addition, firms begin by importing their most important inputs and then expand toward less important ones over time.

Following [Crowley, Meng and Song \(2018\)](#), we estimate the effects of AD investigations on firms’ entry into and exit from sourcing origins for both directly *targeted imports* and indirectly affected *non-targeted imports* within the same firm. For directly targeted imports, our focus is on firms’ *third-country* sourcing responses. When an AD investigation targets a specific product-origin pair, affected firms may respond not only by reducing imports from the investigated origin but also by reallocating sourcing toward or away from other foreign origins supplying the same product. Moreover, as firms devote more resources to targeted imports, they must also adjust their sourcing strategies for non-targeted imports due to binding capacity constraints.

The results show that AD shocks raise the entry probability for targeted imports from third countries by 1.37 percentage points and the corresponding exit probability by 1.53 percentage points. These effects are economically meaningful, given average entry and exit probabilities of 5.44% and 4.07%, respectively. By contrast, for non-targeted imports, firms exposed to AD shocks reduce their entry and exit probabilities by 0.46 and 0.26 percentage points, respectively, relative to average entry and exit rates of 5.43% and 4.57%.

We also find substantial heterogeneity in sourcing capacity across firms: those with broader sourcing networks and greater sourcing capacity are better able to replace suppliers lost because of AD investigations. When exposed to AD shocks, these firms exhibit smaller increases in entry and exit probabilities for targeted imports, as well as smaller declines in entry and exit probabilities for non-targeted imports. These findings support

limited search capacity as an explanation for the evolution of firms' supplier bases and for the interdependence of sourcing decisions across inputs within the firm. More broadly, our analysis highlights the importance of accounting for the effects of a shock on a firm's entire sourcing operation in order to capture the full consequences of supply chain disruptions. Because incumbent products take priority, much of the adjustment cost is borne through reduced search and reallocation for other inputs.

Our findings show that AD investigations reshape firms' broader sourcing networks. The effects are not limited to directly targeted origin-product pairs: firms also adjust third-country sourcing for targeted products and reoptimize sourcing for non-targeted products. The trade effects of AD policy therefore extend beyond directly investigated flows. Therefore, evaluating the aggregate trade effects of AD policy requires a comprehensive understanding of firms' dynamic sourcing adjustments and the substantial churning of foreign supplier relationships they entail. Ignoring cross-product spillovers would potentially understate the overall trade reducing effect of AD policy.

More broadly, these spillovers imply that seemingly natural control groups may be contaminated. Because affected firms adjust not only targeted relationships but also third-country and non-targeted sourcing margins, firms or products that appear untreated may still be indirectly exposed. This underscores the need for caution in defining comparison groups when estimating the effects of trade remedies and other targeted trade-policy shocks.

Armed with our estimates of the effects of AD investigations on firms' sourcing adjustments, we conduct a simple back-of-the-envelope counterfactual exercise to ask what China's imports would have looked like in the absence of AD policy. The results show that aggregate imports would have been higher without AD investigations, through both targeted and non-targeted imports. For targeted imports, in the absence of AD shocks, firms would have been less likely both to search for new origins and to abandon existing ones. The former reduces the creation of new import transactions, whereas the latter supports trade by preserving stable sourcing relationships, so the net effect depends on the relative strength of these two margins. For non-targeted imports, without AD shocks, firms would no longer need to scale back entry into new origins or restrain exit from existing ones. Here too, greater entry raises imports while greater exit lowers them. Overall, the stabilizing effect from less exit on targeted imports dominates the reduction in entry, and for non-targeted imports the positive contribution of additional entry exceeds the negative effect of additional exit. These effects imply higher aggregate import values in the absence of AD investigations.

We contribute to several strands of literature, most directly the literature on antidumping (AD). Empirical AD studies establish two robust patterns: a *trade-depressing* effect on targeted product-country pairs and substantial *trade reallocation* through diver-

sion and deflection. First, AD filings and duties sharply reduce imports of the targeted product from the targeted origin(s), with statistically meaningful trade suppression often detected even when cases are ultimately rejected (Staiger and Wolak, 1994; Krupp and Pollard, 1996; Prusa, 1997, 2001). Second, these bilateral contractions are partly offset by diversion toward non-targeted origins and by deflection of targeted exporters’ sales toward third-country markets (Brenton, 2001; Bown and Crowley, 2006, 2007; Ganguli, 2008; Park, 2009). More recent work examines firm-level adjustments to AD shocks (Konings and Vandenbussche, 2005, 2008; Pierce, 2011; Chandra and Long, 2013) and their aggregate implications (Bown et al., 2021; Zhou, Ruhl and Xu, 2026).

Our analysis is closest to work documenting cross-product adjustments to AD shocks for Chinese firms, notably Crowley, Meng and Song (2018); Lu, Tao and Zhang (2018); Bao, Blonigen and Yu (2026).⁰ This literature primarily studies how exporters adjust once exposed to foreign AD actions. We complement it by focusing on import sourcing responses to China’s domestic AD investigations and by highlighting interdependence in firms’ dynamic adjustments across inputs. We document pronounced *within-product* trade diversion, as firms expand sourcing of targeted inputs toward alternative origins, and a *cross-product* suppression of origin churning, as exposure on some inputs reduces origin churning for other, non-targeted inputs within the same firm. We also find substantial heterogeneity in these adjustments: larger firms respond less along both extensive margins, indicating resource constraints that limit the extent of supplier network adjustments. These novel facts motivate a parsimonious dynamic model of import sourcing with the sourcing capacity constraint.

This paper also speaks to the literature on firms’ global sourcing choices. A central theme in this strand is identifying the determinants of sourcing locations. Firms tend to source an input from origins that offer lower delivered prices, reflecting both origin fundamentals and bilateral trade costs (Eaton and Kortum, 2002; Caliendo and Parro, 2015; Antras, Fort and Tintelnot, 2017). At the same time, they also have incentives to diversify supplier bases and source the same input from multiple locations (Bernard et al., 2018; Blaum, Lelarge and Peters, 2019). Much less is known about how location choices for one input interact with location choices for other inputs, and how this within-firm interdependence evolves dynamically. A notable exception is Lu, Mariscal and Mejía (2024), which studies switching across imported inputs, while abstracting from the origin location margin.

We fill this gap by exploiting policy-driven, input-specific disruptions. Using AD investigations as shocks to narrowly defined product–origin relationships, we document sizable cross-input spillovers in firms’ dynamic sourcing choices, implying that sourcing

⁰See also Lu, Tao and Zhang (2013); Jabbour et al. (2019); Chen et al. (2026) for related evidence on how AD affects Chinese firms’ trade decisions.

adjustments are not separable across inputs in the short run. By relating the magnitude of these spillovers to the breadth of pre-existing supplier bases and to firm size, we further show how product-level network position and firm-level capacity jointly shape the dynamics of global sourcing. In this sense, we complement evidence on dynamic sourcing adjustment following macro shocks (Gopinath and Neiman, 2014; Amiti, Itskhoki and Konings, 2014; Blaum, 2024) by providing micro-level, policy-induced variation that isolates within-firm propagation across inputs.

Last but not least, this paper relates to the work on how production networks transmit idiosyncratic shocks. Existing work has largely emphasized within-country networks. Canonical macro frameworks show how input–output linkages can transform micro shocks into aggregate fluctuations (Acemoglu et al., 2012). Empirically, disruptions hitting domestic suppliers propagate to downstream customers and, through network connections, to other firms, consistent with sizable search and switching costs (Barrot and Sauvagnat, 2016; Carvalho et al., 2016). Conversely, declines in search frictions can facilitate the formation of new buyer–supplier links and improve firm performance (Bernard, Moxnes and Saito, 2019). We contribute by bringing an international sourcing perspective, where buyer–supplier links are plausibly more costly to adjust and where policy shocks can have first-order implications for firms’ input procurement. We show that global sourcing networks transmit disruptions across inputs within the firm: a shock to one set of imported inputs reshapes sourcing adjustments for other, non-targeted inputs. This evidence implies that network-based propagation operates not only through domestic supply chains but also through cross-border buyer–supplier relationships. Our findings also complement a growing body of work that studies global supply-chain frictions using disaggregated data, e.g., Pankratz and Schiller (2024); Grossman, Helpman and Redding (2024) among others.

The remainder of the paper is organized as follows. Section 2 describes the data sources and variable construction. Section 3 provides institutional background on China’s antidumping investigations. Section 4 documents the stylized facts on firms’ foreign supplier bases and sourcing dynamics. Section 5 develops the model and derives testable propositions. Section 6 confronts these predictions with the data, presents the main empirical results, and reports robustness checks. Section 7 conducts back-of-the-envelope counterfactual calculations to quantify the contribution of AD policy to China’s aggregate imports via its impact on extensive-margin sourcing adjustments. Section 8 concludes.

2 Data and Measurements

This section describes the datasets and variable construction used in our empirical analysis. We combine three complementary sources: (i) the Chinese Customs Database, (ii)

the Annual Survey of Industrial Enterprises, and (iii) the Global Antidumping Database. The customs data provide disaggregated firm-level import transactions over time, which we link to production and balance-sheet information from the annual survey. We then merge in product-origin-specific antidumping investigations from the Global Antidumping Database to identify AD shocks and map them to firms’ sourcing activities.

2.1 Data

We obtain detailed import transaction records of Chinese firms from the Chinese Customs Database (CCD) maintained by the General Administration of Customs of China for 2002–2015. We clean and construct the estimation sample in several steps. First, we restrict attention to ordinary import transactions and drop all processing trade transactions, since processing trade is exempt from tariffs and antidumping (AD) duties.¹ Next, we aggregate transactions to the firm–product–origin country–year level. Here, *product* denotes six-digit HS products, harmonized to the 1996 vintage using concordance tables from the United Nations Statistics Division. We then retain only imports of intermediate goods, identified using the Broad Economic Categories classification (BEC Rev. 4), and further restrict attention to manufacturers (i.e., excluding trade intermediaries).² Finally, we restrict the analysis to major trading partners and all origin countries targeted by China’s AD investigations.³ In the final sample, 55 distinct origin countries remain, accounting for more than 90% of China’s total import value. This restriction reduces noise from infrequent sourcing markets and ensures sufficient support for estimating entry and exit dynamics.

This constitutes our full sample of import sourcing by Chinese manufacturing firms over 2002–2015. We use this sample primarily to document the broad patterns of global sourcing, abstracting from AD shocks, and to conduct counterfactual exercises on the effects of AD measures on China’s aggregate import flows based on the estimated impact of AD on firms’ import decisions.

For the firm-level analysis, we further construct a restricted sample to sharpen identification of the AD effects. First, because our focus is on within-firm adjustments across products, we limit the sample to multi-product importers, defined as firms that import at

¹The two main processing trade regimes are pure assembly (PA) and import-and-assembly (IA). Under PA, the ownership of imported inputs remains with the foreign supplier; Chinese firms assemble the inputs and re-export the output to the foreign supplier. Under IA, Chinese firms own the imported inputs but are required to re-export the processed output. In both regimes, firms are exempt from import tariffs and AD duties.

²Following Ahn, Khandelwal and Wei (2011), we identify trade intermediaries as importers whose names contain terms such as “jinchukou,” “maoyi,” and “wuliu.”

³Specifically, we identify the top 20 origin countries by annual import value and take their union over time, which yields 26 distinct countries. Combining this set with the origin countries targeted by China’s AD investigations during 2002–2015 produces 55 origin countries.

least two products in at least one year during the sample period. Second, we retain only firms that import continuously throughout the sample period. This restriction excludes one-off importers and allows us to focus on dynamic sourcing adjustments among active importers. The final dataset for the empirical analysis contains 3,636 Chinese importers, with the median firm sourcing from 4 foreign countries and importing about US\$1.47 million.

The second dataset is the Annual Survey of Industrial Enterprises (ASIE) compiled by the National Bureau of Statistics of China for 1998–2009. It reports firm-level balance-sheet and production information for above-scale manufacturing firms, including domestic sales, employment, and intermediate input purchases. We apply standard sample-cleaning procedures.⁴ Next, we merge ASIE with the customs data using firm identifiers such as company name, address, and phone number. From the combined ASIE–CCD sample, we construct three firm-level size measures: sales revenue, employment, and total import value. Sales revenue and employment are standard measures of production scale from ASIE, while total import value—constructed from CCD—captures the scale of a firm’s foreign sourcing activity and is particularly relevant in our setting. Finally, we augment the data with firm-level revenue-based total factor productivity (RTFP) measures estimated from ASIE. We compute RTFP using three widely used approaches: Olley–Pakes (Olley and Pakes, 1996), Levinsohn–Petrin (Petrin and Levinsohn, 2012), and Akerberg–Caves–Frazer (Akerberg, Caves and Frazer, 2015).

Finally, we obtain the universe of Chinese AD initiations from the Global Antidumping Database (GAD) maintained by the World Bank (Bown, 2016). We then match GAD cases to the CCD and, for each year, identify the product-origin country pairs that were subject to AD investigations in the preceding year.

2.2 Measurements

In this paper, we interpret domestic antidumping (AD) investigations against imported inputs as adverse trade shocks to firms’ foreign supplier relationships. By design, AD measures are specific to targeted products from targeted origin countries and should not directly affect other product-origin pairs.⁵

⁴Specifically, we drop observations with missing values in key accounting variables (total assets, net fixed assets, and gross industrial output), missing employment or employment below 10, negative wage payments, and implausible values for intermediate input use. We then deflate nominal variables using the input and output deflators constructed by Brandt, Van Biesebroeck and Zhang (2012).

⁵For example, an AD investigation against steel from Japan affects imports of Japanese steel, but not steel from the United States or electronics from Japan.

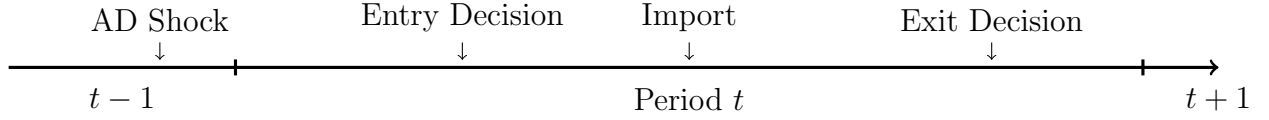


Figure 1: Time Line

Figure 1 summarizes the timing. An AD investigation is realized at the end of period $t - 1$ if any product-origin pair (h, j) in firm f 's import basket becomes subject to AD. At the beginning of period t , the firm may add new origins for its products (entry decisions). It then imports and engages in production during period t . At the end of period t , it may discontinue sourcing from some origins in period $t + 1$ (exit decisions). We therefore examine how AD shocks in $t - 1$ affect entry and exit decisions in t , both defined on period- t observations.

Our main outcomes are entry and exit at the firm-product-origin level. We define *entry* as importing (h, i) in period t but not in $t - 1$, and *exit* as importing (h, i) in period t but not in $t + 1$. Accordingly, $\text{Entry}_{fhi,t} = 1$ if firm f imports (h, i) in t but not in $t - 1$, and $\text{Exit}_{fhi,t} = 1$ if it imports (h, i) in t but not in $t + 1$; both dummies equal zero otherwise. Because the paper studies adjustment of both directly targeted imports and the other non-targeted imports within the same firm, the exact treatment coding used in the regressions is introduced together with the corresponding empirical specifications in Sections 6.2 and 6.3, where we define the treatment indicators for targeted and non-targeted imports and explain the sample restrictions used to isolate direct responses and within-firm spillovers.

3 China's Antidumping Investigations

This section reviews the institutional background and implementation of China's antidumping policy.⁶

3.1 Institutional Background

China introduced its first antidumping (AD) and countervailing legal framework in the late 1990s. The State Council promulgated the *Antidumping and Anti-Subsidy Regulation* in 1997, as part of China's broader effort to align domestic trade-remedy rules with multilateral disciplines ahead of WTO accession. After China joined the WTO, the State Council issued the *Antidumping Regulations of the PRC* in 2002 and revised the regulations again in 2004, reflecting legislative and institutional adjustments during the

⁶See also Messerlin (2004) for a comprehensive review.

early accession period (Wang and Yu, 2002).⁷

Across the 1997, 2002, and 2004 regulations, the Ministry of Foreign Trade and Economic Cooperation (MOFTEC), later reorganized as the Ministry of Commerce (MOFCOM), is the principal body responsible for administering AD investigations, including decisions on initiation and the conduct of investigations.⁸ The State Council’s Customs Tariff Commission is designated in the regulations as the body that issues decisions on the imposition of AD measures following MOFCOM’s proposal.⁹ Customs authorities also play a role in implementation once measures are imposed, including the collection of duties consistent with final determinations.

Under China’s regulations, AD investigations can be initiated following petitions submitted by domestic industry representatives or associations. The regulations also provide for initiation by the competent authority under specified circumstances.¹⁰ Once an investigation is initiated, MOFCOM issues an initiation announcement and proceeds with the investigation in accordance with statutory timelines and procedural steps.

China’s AD regulations specify the core elements of the investigative process broadly aligning with practices in the other countries: the definition of dumping, the assessment of injury, and the requirement of a causal link between dumped imports and injury, together with provisions on questionnaires, verification, hearings, and final determinations. During the early 2000s, aspects of information disclosure and party access to records were discussed in WTO Transitional Review Mechanism reports and related committee documents. Subsequent WTO dispute-settlement reports have addressed specific practices in particular cases, including issues related to disclosure of essential facts and the treatment of confidential information in injury and causation analyses. Interestingly, China’s 1997 regulation includes a provision authorizing the adoption of corresponding measures in response to what the regulation describes as discriminatory use of AD or countervailing measures against China’s exports. This provision was retained in later versions of the regulations, including the 2002 and 2004 revisions.¹¹

These institutional arrangements describe the legal basis and administrative structure under which China’s AD investigations were initiated and implemented during our sample period.

⁷For comparisons across different versions, see Van and Tong (2009). The full texts are available in the official releases of the 1997, 2002, and 2004 regulations.

⁸The regulations specify MOFTEC/MOFCOM’s responsibilities and investigative powers, including the authority to initiate investigations under specified conditions.

⁹WTO Transitional Review Mechanism documents from the early 2000s discuss the institutional allocation of responsibilities in China’s trade-remedy administration.

¹⁰The 1997 regulation includes a provision allowing initiation by the authority after internal consultation when it considers that the evidentiary threshold is met; parallel provisions appear in the 2002 and 2004 regulations.

¹¹The relevant articles appear as Article 40 in the 1997 regulation and as Article 56 in the 2002 and 2004 regulations.

3.2 China's Antidumping Practice

Now we characterize China's AD investigations using the Global Antidumping Database. We focus on two descriptive dimensions: how China's cases are distributed across targeted economies and how investigated incidences are distributed across products.

China's AD practice provides a suitable setting for studying disruptions to firms' foreign supplier bases, because the investigated products are concentrated in HS chapters dominated by raw materials and intermediate inputs. The measures therefore apply directly to the goods firms source for production. We examine the incidences of AD which is defined as HS6–origin country–year triplets. Figure 2 reports AD incidences involving China over 1997–2015. Panel A plots the number of incidences initiated by China. AD activity increased after China's WTO accession in late 2001. During 2001–2006, China initiated 226 investigations against imports from 24 economies, accounting for 58.7% of China's AD initiations over 1997–2015. Panel B plots the number of incidences initiated by foreign authorities against China and China's share in global AD filings. Over the same period, China's share in global AD investigations rose from about 8.7% in 2001 to above 36.9% by 2006.

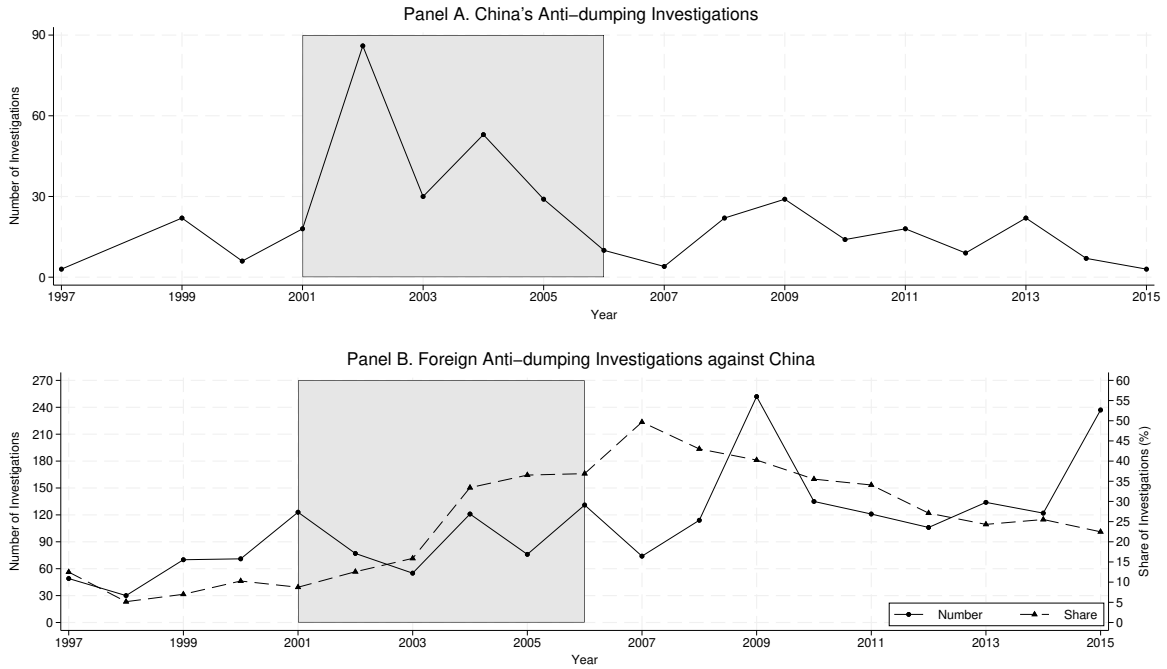


Figure 2: Antidumping Incidences Involving China: Initiations and Being Targeted, 1997–2015

Notes: Antidumping incidences are defined at the HS6–origin–year level. Panel A plots the number of antidumping incidences initiated by China against foreign exporters. Panel B plots the number of antidumping incidences initiated by foreign authorities against China, together with China's share in global AD filings. All series are constructed from the Global Antidumping Database (GAD, 2015) and cover 1997–2015.

Focusing on the post-WTO period from 2001 to 2006, Table 1 reports the distribution of China’s AD incidences across targets and the distribution of foreign AD investigations against China. Panel A aggregates China’s investigations by target economy. South Korea, Japan, Taiwan (China) and the United States together account for 134 incidences (92.0 percent), with Russia also among the frequently named targets. Panel B aggregates foreign investigations against China by filing authority. The United States, the European Union, and India each account for 10–24 percent of the investigations filed against China in this period.

Table 1: Regional Distribution of Antidumping Incidences Involving China, 2001–2006

Panel A. Major Targets of China’s AD Incidences							
Region	South Korea	Japan	Taiwan (China)	USA	Russia	ROW	Total
No. of cases	43	34	29	28	18	74	226
Share (%)	19.0	15.0	12.8	12.4	8.0	32.7	100.0
Panel B. Major Users of AD Incidences against China							
Region	USA	EU	India	Colombia	Canada	ROW	Total
No. of cases	140	82	63	55	50	193	583
Share (%)	24.01	14.1	10.8	9.4	8.6	33.1	100.0

Notes: Antidumping incidences are defined at the HS6–origin–year level. Panel A reports the number and share of China’s AD incidences by target economy, aggregating the five most frequently targeted economies and the rest of the world (ROW). Panel B reports the major users of AD measures against China and their shares in total incidences filed against China.

We next summarize the product coverage of China’s AD investigations during the early post-accession period. These incidences span 11 HS2 sectors and are concentrated in a small number of HS chapters. Table 2 reports the sectoral distribution. Organic chemicals (HS29), paper and paperboard (HS48), and iron and steel (HS72) account for 246 incidences (77.4 percent) over 2001–2006. For reference, Table 2 also reports average MFN tariff changes by HS2 sector over the same period.

Table 2: Product Distribution of China’s Antidumping Incidences: 2001–2006

HS-2 Sector	Title	# of Incidences	Share	MFN Change
29	Organic chemicals	152	47.8%	-2.58%
48	Papers and paperboard, etc.	50	15.7%	-7.23%
72	Iron and steel	44	13.8%	-2.61%
11	Products of the milling industry, etc.	25	7.9%	-13.59%
39	Plastics and articles thereof	13	4.1%	-4.87%
40	Rubber and articles thereof	10	3.2%	-2.43%
54	Man-made filaments, etc.	8	2.5%	-10.41%
38	Miscellaneous chemical products	7	2.2%	-2.56%
28	Inorganic chemicals, etc.	4	1.3%	-2.40%
90	Optical, photographic, etc.	3	0.9%	-3.96%
55	Man-made staple fibers	2	0.6%	-9.90%
Total		318	100%	

Notes: AD incidences are defined at the HS6–origin–year level. MFN Change reports the average MFN tariff change over 2001–2006 within each HS2 sector.

4 Stylized Facts

In this section, we document several facts about Chinese firms’ foreign supplier bases. A foreign supplier base is defined as the set of distinct origin countries from which a firm sources inputs, measured either at the firm level or at the firm–input level. We first establish that firms’ foreign supplier bases are highly active, with substantial origin churning even for a given input in short run. We then show that exposure to AD investigations generates substantial changes in these sourcing adjustments. The magnitude and direction of the response are heterogeneous, varying with whether an import is directly influenced by the shocks and with the breadth of foreign supplier bases. These empirical regularities motivate the model and inform our reduced-form analysis.

4.1 Sourcing Adjustment of Chinese Firms

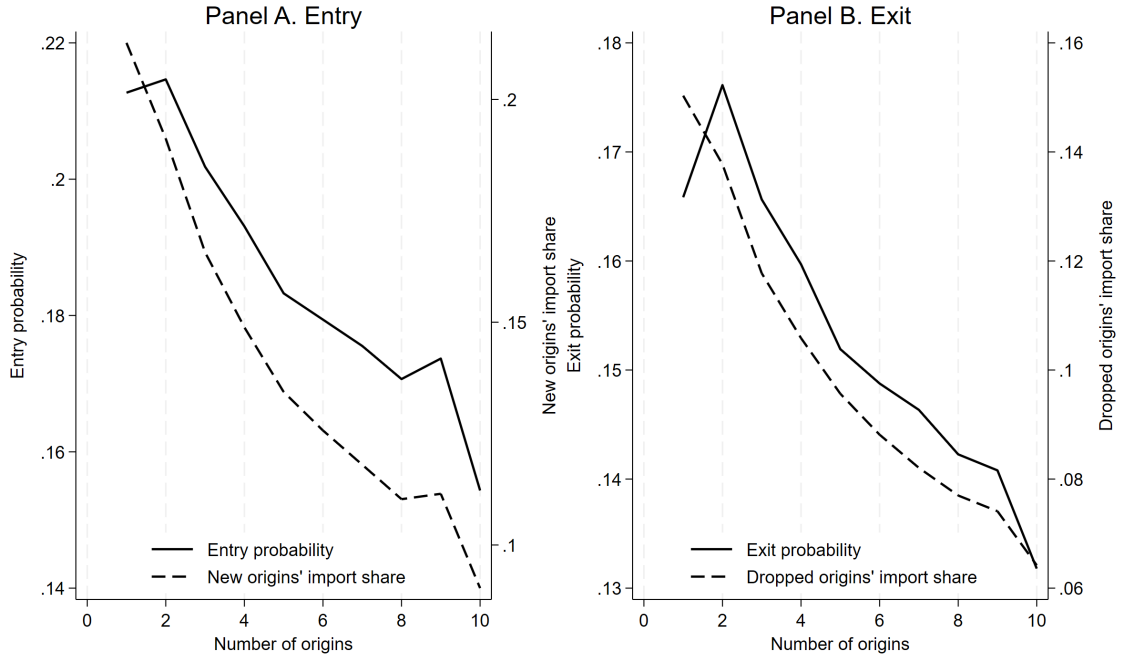
Firms do not simply select a single origin for a given imported input. Instead, they deliberately choose a portfolio of sourcing locations, building sizable foreign supplier bases. Moreover, inputs sourced from multiple foreign countries (i.e., multi-origin imports) are far from marginal. They play a prominent role in shaping both firm-level and aggregate import flows. For an average Chinese importer, multi-origin imports involve about three origins and account for approximately 54.0% of its total import value, while representing only 21.9% of the total number of imported inputs.¹² A similar pattern holds at the

¹²The calculations are performed separately for each firm in each year, followed by averaging across firms and years. This pattern remains stable throughout the sample period. Appendix Figure 6 provides

aggregate level: multi-origin imports account for approximately 80.8% of China’s total import value, yet represent only about 20.3% of firm–product pairs.¹³

More importantly, supplier bases are highly dynamic: firms continually add and drop origins for a given input over time. Figure 3 plots, against the size of the foreign supplier base, the average entry and exit probabilities of sourcing origins and the corresponding import shares accounted for by newly added origins and by origins that will be dropped in the subsequent year. For inputs sourced from more than one origin in a given year, roughly 20% of active origins are new relative to the previous year and account for 12% of current import value. In contrast, 47% of active origins are not used in the subsequent year, and these origins account for 36% of contemporaneous import value. Churning declines as supplier bases expand: firms both experiment less with new origins and shed fewer existing ones. Yet even among inputs sourced from more than 20 origins, turnover remains sizable, with firms adding and dropping about 10% of origins each year. These patterns motivate Stylized Fact 1.

Figure 3: Origin Churning in Chinese Firms’ Foreign Supplier Bases



Notes: Panel A plots the entry probability and the import share of newly added origins against the number of origins for each product. Panel B plots the exit probability and the import share of origins that will be dropped in the following year against the number of origins for each product. Entry (exit) refers to the appearance (disappearance) of an origin relative to the previous (subsequent) year. All variables are defined at the firm–product–year level; the figure reports averages across observations with the same number of origins.

a finer breakdown of the contributions from imports with varying numbers of sourcing origins to firms’ import sourcing activity.

¹³Blaum, Lelarge and Peters (2019) show that sourcing identical HS-8 products from multiple origins is an important driver of firm-level and aggregate import flows in France. We find a comparable pattern for Chinese firms when we use HS-8 data instead of HS-6.

Stylized fact 1. *In China, firm–product pairs that source from multiple foreign origins account for the majority of aggregate imports and exhibit substantial origin churning over time.*

4.2 Sourcing Responses to Antidumping Investigations

Why do firms source the same product from multiple origins and frequently adjust their supplier bases? One explanation is economic: firms may diversify origins to hedge supply risk and preserve the option to reallocate purchases when particular suppliers become less attractive.¹⁴ Consistent with this view, recent studies highlight import churning as an important margin of adjustment to external shocks.

We next examine how firms’ sourcing adjustments differ after exposure to AD investigations. For this descriptive exercise, we classify firms as affected or non-affected. An affected firm is one that, in the year before an AD investigation, imports the investigated product from the investigated origin. All other firms are classified as non-affected.

Within affected firms, we distinguish between directly exposed products and the firm’s remaining products. A directly exposed product is one for which at least one of the firm’s sourcing origins is investigated. The firm’s remaining imported products are referred to as other products. For non-affected firms, we construct the same product grouping only for descriptive comparability. In particular, the comparison group on the “directly exposed” side consists of the same products, but sourced only from origins not named in the AD investigation. Throughout this subsection, therefore, “directly exposed products” should be read as a descriptive label for comparability in Table 3, rather than as the treatment definition used in Section 6.

Table 3 reports the resulting descriptive comparison of entry and exit probabilities. Two patterns stand out. First, for directly exposed products, affected firms exhibit higher entry and exit probabilities than non-affected firms, consistent with rapid reshuffling of sourcing relationships after AD risk rises. Second, for other products, affected firms show lower entry and exit probabilities, suggesting less active adjustment in the rest of the import basket. We summarize these findings in Stylized Fact 2.

Stylized fact 2. *Relative to non-affected firms, affected firms exhibit higher entry and exit for directly exposed products, but lower entry and exit for other products.*

¹⁴A measurement-based explanation is that an HS6 category bundles heterogeneous varieties that are imported from different countries. This is unlikely to drive our results, since we observe similar churning at finer product classifications (HS8).

Table 3: Antidumping Exposure and Firm’s Sourcing Decisions: Descriptive Comparison

		Affected Firms	Non-Affected Firms	Difference
Directly Exposed Products	Entry	9.83% (0.0047)	7.56% (0.0077)	2.27%*** (0.0040)
	Exit	9.01% (0.0047)	7.23% (0.0075)	1.79%** (0.0040)
Other Products	Entry	6.48% (0.0010)	14.15% (0.0021)	-7.67%*** (0.0042)
	Exit	6.91% (0.0009)	12.10% (0.0023)	-5.19%*** (0.0037)

Notes: Parentheses report standard errors of the sample mean. Entry and exit probabilities are computed at the firm–product level and then averaged within each firm–product group across years. The last column reports differences in means (Affected minus Non-Affected); significance is based on two-sided t -tests, with *** ($p < 0.01$), ** ($p < 0.05$), and * ($p < 0.10$). For non-affected firms, the category “Directly Exposed Products” refers to the same product set as in affected firms, but sourced only from non-investigated origins; it is therefore a descriptive comparison category rather than a treated group.

The heightened adjustment for directly exposed products among affected firms is intuitive: AD risk disrupts existing supplier relationships, prompting firms to search for alternative origins (higher entry) and to shift purchases away from some incumbent origins (higher exit). Table 3 also indicates that the entry margin responds somewhat more strongly than the exit margin. By contrast, affected firms adjust less along both margins for other products, suggesting that sourcing for these inputs becomes more stable when firms are simultaneously reconfiguring directly exposed products. This divergence between directly exposed products and other products points to adjustment frictions that constrain firms’ ability to reoptimize all segments of their supplier networks at once.

4.3 Firm Heterogeneity in Sourcing Responses

Finally, we examine heterogeneity in firms’ responses to determine whether adjustment patterns vary with the breadth of their pre-existing supplier base. Firms with broader foreign sourcing networks may be better able to reallocate purchases across established origins, whereas firms with narrower networks may need to search more aggressively and therefore exhibit greater churn following a shock. This reasoning suggests that sourcing adjustments should be more sensitive to AD shocks when a firm’s supplier base is narrower.

Table 4 examines this implication by classifying affected firms using two proxies for breadth of supplier base: the number of foreign sourcing origins and the average distance to those origins.¹⁵ Firms sourcing from a larger set of countries and/or from more distant origins are interpreted as having broader supplier bases. Specifically, we define the group

¹⁵The average distance to a firm’s sourcing origins is calculated as the simple average of the bilateral geographic distances between China and each origin country from which the firm sources. The distance data are drawn from the CEPII Gravity Dataset.

of large firms as firms with above-median values for both the number of origin countries and the average sourcing distance. The results indicate that firms with narrower supplier bases respond more strongly to AD shocks, exhibiting larger increases in both entry and exit. We summarize this pattern in Stylized Fact 3.

Table 4: Antidumping Shocks and Firms’ Sourcing Decisions: Supplier-Base Heterogeneity

		# of Origins			Average Distance		
		Big	Small	Difference	Big	Small	Difference
Targeted Imports	Entry	9.54% (0.0076)	20.31% (0.0705)	-10.78%*** (0.0473)	9.34% (0.0090)	10.85% (0.0143)	-1.51% (0.0164)
	Exit	8.78% (0.0074)	17.19% (0.0659)	-8.41%*** (0.0457)	8.65% (0.0089)	9.76% (0.0137)	-1.11%** (0.0159)
Non-Targeted Imports	Entry	6.48% (0.0021)	6.40% (0.0090)	0.09% (0.0067)	6.79% (0.0028)	6.05% (0.0033)	0.75%** (0.0043)
	Exit	7.31% (0.0025)	3.73% (0.0059)	3.58%*** (0.0072)	7.67% (0.0030)	5.88% (0.0034)	1.79%*** (0.0046)

Notes: Parentheses report standard errors of the sample mean. Entry and exit probabilities are first computed at the firm–product level and then averaged within groups defined by firm type (big vs. small) and import type (targeted vs. non-targeted). The last column reports differences in means (big minus small) along with t -tests for equality of means; ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

Stylized fact 3. *Among affected firms, the sourcing response to AD shocks is markedly stronger for those with narrower foreign supplier bases. Relative to firms with broader sourcing networks, firms importing from fewer origins or from geographically closer countries display higher probabilities of entry and exit for targeted imports, but less entry and exit probabilities for non-targeted imports.*

5 A Simple Model of Sourcing Capacity

In this section, we develop and analyze a simple model of import sourcing that rationalizes the key empirical regularities documented above. The model is intentionally parsimonious, but it highlights two central forces shaping firms’ dynamic sourcing decisions. First, firms benefit from expanding their foreign supplier bases through lower marginal costs and improved input availability. Second, building and reshaping a supplier base is costly, as firms face resource constraints in expanding sourcing relationships. These forces generate endogenous supplier-base dynamics and yield testable predictions that guide our reduced-form analysis of AD shocks.

5.1 Technology

Consider a firm that produces a final good using a set H of composite intermediate inputs indexed by h . For each h , the firm combines a continuum of input varieties indexed by

$\omega \in [0, M]$ using a Cobb–Douglas aggregator,

$$c_{h,t} = \exp\left(\int_0^M \ln c_{h,t}(\omega) d\omega\right), \quad (1)$$

where c_{ht} denotes the unit cost of producing composite input h at time t and $c_{h,t}(\omega)$ is the unit cost of variety ω .

Each variety can be sourced either domestically at normalized unit cost 1, or from abroad at unit cost $c < 1$. Let $x_{h,t} \in [0, M]$ denote the measure of varieties of input h that the firm sources from foreign suppliers at time t . Then (1) implies

$$c_{h,t} = \exp(x_{h,t} \ln c), \quad (2)$$

so that a larger foreign supplier base lowers the unit cost of the composite input. Intuitively, because $\ln c < 0$, expanding $x_{h,t}$ reduces $c_{h,t}$ proportionally in logs.

Dynamics and sourcing. We introduce supplier dynamics and costly search. Let $x_{h,t}$ be the stock (measure) of foreign suppliers the firm has identified for input h at time t . Each period, existing foreign sourcing relationships break down with hazard rate $\mu = \mu(x_{h,t}) \in (0, 1)$, capturing supplier exit, contract termination, or disruptions. We assume that $\mu(x_{h,t})$ is increasing in the current stock of foreign suppliers, capturing the notion that managing a broader supplier network becomes increasingly difficult for firms. The firm can also expand its foreign supplier base by investing sourcing effort $n_{h,t} \geq 0$ (interpreted as search intensity or managerial attention allocated to input h). The law of motion is

$$x_{h,t+1} = x_{h,t}(1 - \mu) + n_{h,t}, \quad (3)$$

so that next period's supplier base equals the surviving relationships plus newly established ones. Combining (2) and (3), the unit cost of composite input h evolves as

$$c_{h,t+1} = \exp(\ln c [x_{h,t}(1 - \mu) + n_{h,t}]). \quad (4)$$

Final-good technology. The firm assembles the final good from the composite inputs using a Leontief technology,

$$c_t = \sum_{h \in H} \phi_h c_{h,t}, \quad (5)$$

where $\phi_h > 0$ is the input requirement (or cost share weight) for composite input h . This structure highlights that improvements in any composite input translate one-for-one into lower unit cost of final output, with intensity governed by ϕ_h .

Sourcing capacity. At any point in time, the firm is characterized by its vector of supplier stocks $\{x_{h,t}\}_{h \in H}$ and a limited sourcing capacity $\bar{n} > 0$ that must be allocated across inputs:

$$\sum_{h \in H} n_{h,t} \leq \bar{n}. \quad (6)$$

This constraint captures the idea that supplier discovery and relationship management require scarce managerial resources. As a result, the firm may concentrate search on a subset of inputs in a given period, leaving other inputs with little or no search investment.

5.2 Optimal Searching and Import Adjustment

The firm allocates scarce sourcing capacity across inputs to minimize current unit costs. The optimal allocation depends on (i) the existing stock of foreign suppliers for each input, (ii) the firm's total sourcing capacity, and (iii) the importance of each input in production, as captured by its cost share. Because search intensity cannot be negative, the problem features corner solutions and naturally generates a set of inputs that receive positive search investment in a given period.

Static problem. Taking the state vector $\{x_{h,t}\}_{h \in H}$ as given, the firm chooses $\{n_{h,t}\}_{h \in H}$ to minimize the unit cost of production subject to the capacity constraint $\sum_h n_{h,t} \leq \bar{n}$ and non-negativity $n_{h,t} \geq 0$:

$$\begin{aligned} \min_{\{n_{h,t} \geq 0\}} \quad & \sum_{h \in H} \phi_h \exp(\ln c [x_{h,t}(1 - \mu) + n_{h,t}]) \\ \text{s.t.} \quad & \sum_{h \in H} n_{h,t} \leq \bar{n}. \end{aligned} \quad (7)$$

Let $\Phi_t \geq 0$ denote the multiplier on the capacity constraint and $\lambda_{h,t} \geq 0$ the multipliers on $n_{h,t} \geq 0$. The Lagrangian is

$$\mathcal{L}_t = \sum_{h \in H} \phi_h \exp(\ln c [x_{h,t}(1 - \mu) + n_{h,t}]) - \Phi_t \left(\bar{n} - \sum_{h \in H} n_{h,t} \right) - \sum_{h \in H} \lambda_{h,t} n_{h,t}. \quad (8)$$

The first-order condition for $n_{h,t}$ is

$$-(\ln c) \phi_h \exp(\ln c [x_{h,t}(1 - \mu) + n_{h,t}]) = \Phi_t - \lambda_{h,t}. \quad (9)$$

Since $\ln c < 0$, the left-hand side is positive and can be interpreted as the marginal cost saving from allocating one more unit of search to input h .

Funded inputs and interior solution. Let $H_t \subseteq H$ denote the set of *funded* inputs with $n_{h,t} > 0$ at time t . For $h \in H_t$, complementary slackness implies $\lambda_{h,t} = 0$, and (9) can be solved for the interior allocation:

$$n_{h,t} = \frac{1}{|\ln c|} \left[\ln \phi_h - \ln \left(\frac{\Phi_t}{|\ln c|} \right) \right] - (1 - \mu)x_{h,t}. \quad (10)$$

Thus, conditional on being funded, search increases with the input's importance (higher ϕ_h) and decreases with the surviving stock of suppliers (higher $(1 - \mu)x_{h,t}$). If $x_{h,t}$ is sufficiently large, the right-hand side becomes negative and the corner $n_{h,t} = 0$ is optimal.

Summing (10) over $h \in H_t$ and imposing the capacity constraint yields an expression that pins down the shadow value of sourcing capacity:

$$\frac{1}{|\ln c|} \ln \left(\frac{\Phi_t}{|\ln c|} \right) = \frac{1}{|H_t|} \left(\sum_{h \in H_t} \frac{\ln \phi_h}{|\ln c|} - (1 - \mu) \sum_{h \in H_t} x_{h,t} - \bar{n} \right), \quad (11)$$

where $|H_t|$ is the number of funded inputs.

Lemma 1. *The shadow price of sourcing capacity, Φ_t , is decreasing in total sourcing capacity $\bar{n}$ and in the average surviving stock of supplier contacts among funded inputs, $\frac{1}{|H_t|} \sum_{h \in H_t} x_{h,t}$.*

Allocation across funded inputs. Combining (10) and (11) yields a transparent decomposition:

$$n_{h,t} = \frac{\bar{n}}{|H_t|} + \left(\frac{\ln \phi_h}{|\ln c|} - (1 - \mu)x_{h,t} \right) - \frac{1}{|H_t|} \sum_{k \in H_t} \left(\frac{\ln \phi_k}{|\ln c|} - (1 - \mu)x_{k,t} \right). \quad (12)$$

Absent asymmetries, the firm spreads capacity evenly across funded inputs, $\bar{n}/|H_t|$. Asymmetries arise because some inputs matter more (high ϕ_h) and because some inputs already have a richer supplier base (high $x_{h,t}$), both relative to funded averages.

Proposition 1. *Holding fixed the set of funded inputs H_t , the number of newly added foreign suppliers is higher for inputs with (i) a smaller existing supplier base (lower $x_{h,t}$) and (ii) a higher cost share (higher ϕ_h). Moreover, if $x_{h,t}$ is sufficiently large, the corner solution binds and the firm sets $n_{h,t} = 0$, ceasing search for input h .*

Dynamics of the shadow value. To study how the shadow value evolves, hold the set of funded inputs fixed at H_t and difference (11) over time:

$$\ln \Phi_{t+1} - \ln \Phi_t = -(1 - \mu) \frac{|\ln c|}{|H_t|} \sum_{h \in H_t} (x_{h,t+1} - x_{h,t}). \quad (13)$$

Using the law of motion $x_{h,t+1} = x_{h,t}(1 - \mu) + n_{h,t}$ and summing over funded inputs gives

$$\ln \Phi_{t+1} - \ln \Phi_t = -(1 - \mu) \frac{|\ln c|}{|H_t|} \left(\bar{n} - \mu \sum_{h \in H_t} x_{h,t} \right). \quad (14)$$

Hence, when funded supplier stocks are low (so that $\mu \sum_{h \in H_t} x_{h,t} < \bar{n}$), the shadow price declines over time as the firm accumulates contacts. In a stationary configuration with a fixed funded set, the average inflow of new suppliers per funded input equals the depreciation of the existing supplier stock, implying

$$\frac{1}{|\ln c|} \ln \left(\frac{\Phi_t}{|\ln c|} \right) = \frac{1}{|H_t|} \sum_{h \in H_t} \frac{\ln \phi_h}{|\ln c|} - \frac{1}{|H_t|} \frac{\bar{n}}{\mu}. \quad (15)$$

Which inputs are funded? For an unfunded input $i \notin H_t$, the marginal benefit of initiating search must be below the shadow cost:

$$\ln \phi_i + \ln c(x_{i,t}(1 - \mu)) < \ln \left(\frac{\Phi_t}{|\ln c|} \right). \quad (16)$$

Starting from scratch ($x_{i,t} = 0$), this reduces to a cutoff rule: an input is funded only if its importance ϕ_i is sufficiently high relative to the shadow cost of capacity. Ordering inputs by ϕ_i , a steady state with $|H_t|$ funded inputs exists if

$$\max_{i \notin H_t} \ln \phi_i < \frac{1}{|H_t|} \left(\sum_{h \in H_t} \ln \phi_h - |\ln c| \frac{\bar{n}}{\mu} \right). \quad (17)$$

When this condition fails, the funded set must expand to include additional (less important) inputs.

Proposition 2. *Firms with greater sourcing capacity $\bar{n}$ and/or a larger accumulated stock of supplier contacts $\sum_{h \in H} x_{h,t}$ import a wider range of inputs. A firm that is new to importing begins by searching for the most important inputs (high ϕ_h) and expands over time toward less important ones.*

Proposition 2 links import experience to both breadth and dynamics: as firms accumulate supplier contacts, they expand their imported input scope and reduce unit costs, while continuing to exhibit ongoing churn as relationships depreciate and are replenished.

5.3 Comparative Statics: A Negative Shock

We now study how a firm reallocates search effort after an adverse shock to its supplier base. Consider an unexpected disruption that reduces the stock of foreign supplier

contacts for a particular input h at time t (e.g., the loss of a key origin, a relationship breakdown, or a policy shock that makes a subset of existing suppliers infeasible). Formally, holding fixed the firm’s total capacity $\bar{n}$, the shock lowers $x_{h,t}$.

Direct effect on the shocked input. For a funded input, the optimal search allocation (10) implies that, holding the shadow price Φ_t fixed, a decline in $x_{h,t}$ increases search effort for the shocked input:

$$\frac{\partial n_{h,t}}{\partial x_{h,t}} = -(1 - \mu) < 0.$$

Intuitively, when the firm loses contacts in input h , the marginal return to rebuilding that supplier base rises, so the firm allocates more of its limited search capacity to that input. This mechanism generates a *replacement motive*: following a disruption, the firm actively searches for new origins for the shocked input.

Effects through the shadow price. The shock also affects the shadow price of sourcing capacity. Equation (11) shows that Φ_t is increasing in the (average) scarcity of effective supplier contacts among funded inputs. When $x_{h,t}$ falls for a funded input, the term $(1 - \mu) \sum_{k \in H_t} x_{k,t}$ declines, raising $\ln(\Phi_t / |\ln c|)$ and hence Φ_t . This increase in the shadow price captures the fact that capacity becomes more valuable when the firm’s existing supplier base is thinner: every unit of search can now generate larger cost reductions.

A higher Φ_t has two implications. First, it reduces search allocations for all *other* funded inputs through (10), because the marginal benefit of search must rise to justify diverting scarce capacity away from rebuilding the disrupted input. Second, if the increase in Φ_t is sufficiently large, it can push some marginal inputs to the corner, $n_{i,t} = 0$, shrinking the set of funded inputs H_t . This is the key *extensive-margin* adjustment: the firm may temporarily stop searching in some inputs to free up capacity for the disrupted one.

Targeted versus non-targeted adjustments. The model delivers a natural asymmetry across inputs after a disruption. The shocked input h experiences higher search intensity (more entry into new origins). In contrast, other inputs see lower search intensity and may even be defunded (less entry). Moreover, because lower entry mechanically reduces the scope for later pruning, exit responses can diverge as well: the shocked input can exhibit both higher entry and higher exit as the firm experiments, overshoots, and subsequently rationalizes its supplier base (recall that $\mu(x_{h,t})$ is increasing in current supplier stock $x_{h,t}$); non-shocked inputs can exhibit lower entry and lower exit due to reduced experimentation and a greater incentive to preserve existing relationships when future re-entry is costly.

Firm size and capacity. The same logic implies systematic heterogeneity across firms. Firms with larger sourcing capacity $\bar{n}$ face a lower shadow price of search in normal times (Lemma 1) and can respond to disruptions with less crowding out. After a given absolute shock to $x_{h,t}$, a large-capacity firm can increase n_{ht} while maintaining positive search in other inputs, implying a smaller contraction in $|H_t|$ and a weaker reduction in search for non-shocked inputs. By contrast, a small-capacity firm experiences a larger increase in the shadow price Φ_t for the same disruption and therefore reallocates more aggressively toward the shocked input, with stronger crowding out elsewhere. This mechanism suggests that smaller firms (or firms with narrower supplier bases) should display larger extensive-margin responses to negative shocks.

These comparative statics directly motivate our empirical analysis: we test whether AD exposure (i) increases entry and exit for the targeted product as firms rebuild and reoptimize their supplier bases, (ii) reduces entry and exit for non-targeted products due to capacity reallocation, and (iii) generates stronger responses among smaller firms and firms with narrower pre-shock supplier bases.

6 Empirical Analysis

In this section, we confront the model with the data and conduct reduced-form tests of Chinese firms' import sourcing adjustments. We proceed in several steps. Section 6.1 validates the model's key mechanisms. Sections 6.2 and 6.3 focus on sourcing responses to supply chain disruptions and explore heterogeneity across firms. Section 6.4 reports robustness checks. Section 6.5 provides a detailed discussion of how our empirical strategies connect to the theoretical framework. Overall, the evidence closely matches the stylized facts and the model's predictions.

We define *targeted imports* as firm-product pairs at period t in which (i) firm f imported HS-6 product h from some source country i at $t - 1$, and (ii) China initiated an investigation against (h, i) at $t - 1$. *Non-targeted imports* refer to all other firm-product pairs where firms are not directly subject to such trade disruptions. Before turning to the regressions, it is useful to clarify how the empirical comparisons map into the two margins of adjustment highlighted by the model. The descriptive evidence in Section 4 compares affected and non-affected firms to establish the basic patterns in the data. The regression analysis then sharpens identification in two different ways. For targeted imports, we use firm-product-level exposure and firm-year fixed effects to isolate the direct response of the product hit by the AD shock. For non-targeted imports, we exclude directly targeted products and use firm-level exposure to isolate spillovers to the rest of the firm's import basket. The two designs are therefore intentionally different because they are aimed at different objectives.

6.1 Validating Model Mechanisms

Propositions 1 and 2 deliver testable predictions about firms’ sourcing adjustments. We measure the number of newly added foreign suppliers for firm f and input h in period t as the number of newly added sourcing origins from which the firm imports h at t relative to $t - 1$. To proxy the breadth of the foreign supplier base at the firm–input level, we use (i) the number of sourcing origins and (ii) the average distance from sourcing origins to the destination country (China). At the firm level, we measure supplier-base breadth as the number of sourcing origins across all imported inputs; this also serves as a proxy for the firm’s search capacity.

Proposition 1 predicts that the number of newly added suppliers for an input is decreasing in the size of its established supplier base and increasing in its import cost share. Table 5 supports these predictions: firms add more origins for inputs with narrower supplier bases and for inputs with higher import cost shares. Both propositions receive empirical support and align well with the observed patterns of sourcing adjustment.

Table 5: Churning, Supplier Base, and Import Cost Share

	Net Number of Added Origins $_{fh,t+1}$				
	(1)	(2)	(3)	(4)	(5)
# of Origins $_{fh,t}$	-0.563*** (0.00366)			-0.603*** (0.00356)	
Avg. Distance of Origins $_{fh,t}$		-0.0893*** (0.00160)			-0.0906*** (0.00159)
Share $_{fh,t}$			0.486*** (0.00854)	0.939*** (0.0100)	0.491*** (0.00854)
Firm FE	Yes	Yes	Yes	Yes	Yes
Product FE	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes
Observations	1,723,051	1,723,051	1,723,409	1,723,051	1,723,051
Adj. R ²	0.0887	0.0419	0.0424	0.0962	0.0441

Note: Standard errors are reported in parentheses and clustered at the firm–country level. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

Proposition 2 links a firm’s search capacity to its import scope and the evolution of imported inputs over time. Table 6, columns (1) and (2), shows that firms with broader supplier bases (and thus greater search capacity) import a wider set of inputs. Columns (3) and (4) further indicate that newly imported inputs—those with lower import age—tend to have lower cost shares. That is, firms begin by importing their most important inputs and subsequently expand toward less important ones.

Table 6: Firms' Import Strategy

	# of Imported Products _{<i>f,t</i>}		Import Cost Share _{<i>f,h,t</i>}	
	(1)	(2)	(3)	(4)
log # of Origins _{<i>f,t</i>}	0.899*** (0.002)			
log Avg. Distance of Origins _{<i>f,t</i>}		0.231*** (0.003)		
Import Age _{<i>f,h,t</i>}			0.029*** (0.001)	
$\mathbb{1}\{\text{New Import}\}_{f,h,t}$				-0.028*** (0.001)
Firm FE	Yes	Yes	Yes	Yes
Product FE	-	-	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Observations	1,056,621	1,056,621	1,723,409	1,723,409
Adjusted R ²	0.707	0.707	0.253	0.253

Note: Standard errors, reported in parentheses, are clustered at the firm level in columns (1) and (2), and at the firm-product level in columns (3) and (4). ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

6.2 Sourcing Adjustment of Targeted Imports

We move on to examine how AD investigations affect entry into and exit from origin countries for *targeted imports*. Our objective is to examine the direct reallocation of sourcing effort toward a specific product after some of its sourcing origins are hit by AD shocks, conditional on the firm continuing to import that product. Throughout this subsection, the focus is on *third-country* responses, that is, adjustments in origins other than the origin(s) directly targeted by the AD investigation.

Following [Crowley, Meng and Song \(2018\)](#), we estimate the following linear probability specification for both entry and exit:

$$\text{Entry}_{fhi,t} \text{ or } \text{Exit}_{fhi,t} = \alpha + \beta_1 \text{AD}_{fh,t-1} + \delta_{fh} + \eta_{f,t} + \omega_{hc,t} + \epsilon_{fhi,t}, \quad (18)$$

where $\text{Entry}_{fhi,t}$ equals one if firm f starts importing HS6 product h from origin i in year t , and zero otherwise; $\text{Exit}_{fhi,t}$ equals one if firm f stops importing HS6 product h from origin i in year $t + 1$, that is, if the origin is dropped after year t , and zero otherwise. We include firm-product fixed effects δ_{fh} , firm-year fixed effects $\eta_{f,t}$, and product-origin-year fixed effects $\omega_{hc,t}$. Standard errors are clustered at the firm-product level.

We retain both targeted and non-targeted imports in the estimation sample and define treatment at the firm-product level through $\text{AD}_{fh,t-1}$. Specifically, $\text{AD}_{fh,t-1} = 1$ if firm f imported product h from at least one origin targeted by an AD investigation in period $t - 1$ and continues to import h in period t , and $\text{AD}_{fh,t-1} = 0$ otherwise. Treatment is therefore defined at the firm-product level rather than the firm level, so a firm's other products

remain in the estimation sample even when only one product is directly exposed to the AD shock. The continuation requirement is important. The targeted-import regressions are designed to study supplier-base reconfiguration for a product that remains in the firm’s import basket after the shock. If the firm imported h in $t - 1$ but no longer imports h at all in t , there is no current-year origin margin left to observe for that product, and including such cases would conflate within-product origin reallocation with product-level import exit. The key comparison in this specification is therefore not simply between affected and non-affected firms. Instead, it is between directly exposed firm-product pairs and the same firm’s other products in the same year, while also comparing against other firms sourcing in the same product-origin-year market.¹⁶ Appendix A.1 provides additional details on the construction of the key variables and the estimation sample, and Appendix A.2 presents the summary statistics.

Our empirical design focuses on *third-country* responses for targeted imports. The idea is that an AD investigation directly affects imports from the investigated origin, but the margin of interest here is how firms reconfigure the rest of the supplier base for the same product after that disruption. We therefore exclude directly targeted origins from the estimation sample and restrict attention to third-country origins. This ensures that the estimates capture supplier-base reconfiguration after the shock, rather than the mechanical contraction of imports from the investigated origin itself.¹⁷

The timing of entry and exit follows directly from this setup. Entry is defined backward-looking: an origin counts as an entry in year t if it is used in t but not in $t - 1$. Exit is defined forward-looking: an origin counts as an exit in year t if it is used in t but not in $t + 1$. Thus, both outcomes are centered on origins that are active in year t , with entry capturing newly added relationships and exit capturing relationships that are about to be discontinued. The asymmetry in timing is therefore mechanical rather than conceptual.

In addition, if a product-origin pair (h', i) becomes targeted in period t , we drop (f, h', i) in t so that the estimated response is interpreted as a one-period-lagged response to exposure realized in $t - 1$. One feature of the data makes this margin economically meaningful. For targeted products, imports from the targeted origin account for 84.7% of a firm’s total import value of targeted imports, on average. AD investigations therefore typically generate a sizable shock to targeted imports, so meaningful supplier-base adjustment should be expected. A coding illustration is provided in the appendix.

¹⁶This distinction is crucial because one of the central lessons of our analysis is that AD investigations generate spillovers across sourcing relationships within the firm. A comparison group that includes indirectly affected firm-product-origin relationships may therefore understate the full effect of AD exposure.

¹⁷If directly targeted origins were included in the estimation sample, the estimated effect would partly reflect the standard trade-depressing effect of AD on the investigated origin itself, which is conceptually distinct from the supplier-base reconfiguration that is the focus of our analysis.

We exploit variation in exposure at the firm-product level and estimate entry and exit at the firm-product-origin level. Firm-product fixed effects absorb time-invariant differences across firm-product pairs, such as a firm’s baseline reliance on a particular product or its persistent tendency to add and drop origins. Firm-year fixed effects absorb all shocks common to the firm in a given year, including firm-wide changes in demand, scale, and financing conditions. Product-origin-year fixed effects absorb shocks common to a sourcing market in a given year, such as changes in foreign supply conditions, bilateral trade costs, or other product-origin-specific disturbances. Importantly, any firm-level response to AD that is common across the entire import basket is also absorbed by these firm-year fixed effects.

The coefficient β_1 therefore captures how a firm adjusts sourcing for directly exposed products relative to its other products in the same year, relative to the same firm-product pair in years without exposure, and relative to other firms sourcing in the same product-origin market in the same year. The identifying assumption is that, conditional on these fixed effects, the timing of AD exposure is not correlated with other unobserved shocks that differentially affect entry or exit for the directly exposed imports. Under this assumption, the specification isolates the direct product-specific effect of AD exposure on sourcing adjustments for targeted imports, rather than broader firm-level changes in import behavior.

We expect $\beta_1 > 0$ for both entry and exit. AD exposure should induce firms to add new origins for the targeted product while also increasing the likelihood of dropping existing ones. In the model, these two responses arise naturally from supplier network reconfiguration: firms engage in compensatory search for alternative origins after the shock (i.e., an increase in $n_{h,t}$), but the resulting expansion of the supplier base also makes global sourcing relationships harder to manage, leading to more exit from existing origins (i.e., an increase in $\mu_{ht}(n_{h,t})$).

Finding 1. *AD exposure increases both entry into new third-country origins and exit from established third-country origins for targeted imports. In addition, the responses are stronger for firms with limited sourcing capacity measured by import value and number of sourcing origins.*

Table 7 reports the baseline estimates. Columns (1)–(2) present entry results and columns (3)–(4) present exit results. Columns (2) and (4) restrict the sample to existing products, excluding new products.¹⁸ The coefficient on $AD_{fh,t-1}$ is positive and statistically significant throughout. Quantitatively, AD exposure increases the probability of entering a new third-country origin by 1.37 percentage points in the full sample (column

¹⁸Because targeted imports must be existing products in the firm’s import basket, newly added products necessarily have $AD_{fh,t-1} = 0$. Excluding new products therefore sharpens identification by removing observations that are mechanically untreated.

(1)) and by 1.80 percentage points among existing products (column (2)), and increases the probability of dropping an established origin by 1.53 percentage points in the full sample (column (3)) and 1.30 percentage points among existing products (column (4)).

Table 7: Entry and Exit Decisions for Targeted Imports from Third Countries

	Entry _{fhc,t}		Exit _{fhc,t}	
	(1)	(2)	(3)	(4)
	All Products	Existing Products	All Products	Existing Products
AD _{fh,t}	0.0137** (0.00595)	0.0180*** (0.00576)	0.0153*** (0.00558)	0.0130** (0.00590)
Firm-Product FE	YES	YES	YES	YES
Firm-Year FE	YES	YES	YES	YES
Product-Origin-Year FE	YES	YES	YES	YES
Mean Prob.	5.44%	2.90%	4.07%	4.03%
AD Contribution	25.18%	62.07%	37.59%	32.26%
Obs.	2,158,187	1,359,959	2,001,753	1,359,959
Adj. R ²	0.360	0.226	0.360	0.349

Notes: We focus on firms' import-sourcing responses for targeted imports from third-country origins, namely origins that supply targeted products but are not investigated by China. Standard errors clustered at the firm-country level are reported in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively. "All products" includes both products that the firm did not import in the previous year but begins importing in the current year (i.e., new products) and products that the firm continues to import from the previous year (i.e., existing products). AD contribution is calculated as the estimated coefficient of AD dummy in each specification divided by the corresponding mean probability.

These effects are sizable among affected firms. The average entry probability is 5.44% for all products and 2.90% for existing products, implying that AD exposure raises entry by 25.18% and 62.07%, respectively. The average exit probability is 4.07% for all products and 4.03% for existing products, implying relative increases of 37.59% and 32.26%.

We then explore heterogeneity by firms' sourcing capacity. According to our theory, firms with greater sourcing capacity should be better able to absorb disruptions to targeted imports and thus adjust less on the extensive margins. To test this prediction, we augment the baseline specification with a capacity measure and its interaction with AD exposure:

$$\begin{aligned} \text{Entry}_{fhi,t} \text{ or } \text{Exit}_{fhi,t} = & \beta_1 \text{AD}_{fh,t-1} + \beta_2 (\text{AD}_{fh,t-1} \times \text{Capacity}_{f,t-1}) \\ & + \delta_{fh} + \eta_{f,t} + \omega_{hc,t} + \epsilon_{fhi,t}. \end{aligned} \quad (19)$$

Here, $\text{Capacity}_{f,t-1}$ is a firm-level capacity proxy measured at $t-1$. In the baseline specification, we proxy firms' sourcing capacity using three measures: log import value, the log number of distinct sourcing origins, and the average distance between China and those sourcing origins. The interaction term β_2 captures how the AD response varies with firm capacity. We expect $\beta_2 < 0$: AD exposure increases both origin entry and origin exit, but these responses are systematically weaker for larger firms.

Table 8 reports the results from (19) using three alternative proxies for sourcing capacity. Across all specifications, the coefficient on $AD_{fh,t-1}$ remains positive and statistically significant, while the interaction term $AD_{fh,t-1} \times Capacity_{f,t-1}$ is negative and statistically significant. The only exception is the specification using geographic distance as a proxy, which yields coefficients with the expected signs but without statistical significance. These results suggest that AD exposure induces sourcing reconfiguration for all firms, but the response is stronger among firms with more limited sourcing capacity, such as those with smaller import values, fewer sourcing origins, or sourcing relationships concentrated in nearby countries. We summarize the responses of targeted imports in Finding 1.

Table 8: Heterogeneous Responses of Targeted Imports from Third Countries

Panel A: Entry Decisions					
	All Products			Existing Products	
	(1)	(2)	(3)	(4)	(6)
$AD_{f,h,t}$	0.107** (0.0416)	0.0437*** (0.0104)	0.0822 (0.193)	0.101** (0.0420)	0.0494*** (0.0107)
$\times \log \text{ Import Value}_{f,t-1}$	-0.00548** (0.00231)			-0.00490** (0.00232)	
$\times \log \text{ Number of Origins}_{f,t-1}$		-0.0264*** (0.00711)			-0.0287*** (0.00757)
$\times \log \text{ Average Distance of Origins}_{f,t-1}$			-0.00786 (0.0219)		-0.0178 (0.0226)
Firm-Product FE	YES	YES	YES	YES	YES
Firm-Year FE	YES	YES	YES	YES	YES
Product-Origin-Year FE	YES	YES	YES	YES	YES
Obs.	2,057,364	2,057,364	2,057,361	1,313,264	1,313,264
Adj. R ²	0.313	0.313	0.313	0.223	0.223
Panel B: Exit Decisions					
	All Products			Existing Products	
	(1)	(2)	(3)	(4)	(6)
$AD_{f,h,t}$	0.0998*** (0.0372)	0.0345*** (0.00982)	0.341* (0.177)	0.0845** (0.0371)	0.0314*** (0.0103)
$\times \log \text{ Import Value}_{f,t-1}$	-0.00491** (0.00209)			-0.00412** (0.00206)	
$\times \log \text{ Number of Origins}_{f,t-1}$		-0.0185*** (0.00673)			-0.0166** (0.00706)
$\times \log \text{ Average Distance of Origins}_{f,t-1}$			-0.0369* (0.0200)		-0.0272 (0.0203)
Firm-Product FE	YES	YES	YES	YES	YES
Firm-Year FE	YES	YES	YES	YES	YES
Product-Origin-Year FE	YES	YES	YES	YES	YES
Obs.	1,900,974	1,900,974	1,900,971	1,313,264	1,313,264
Adj. R ²	0.362	0.362	0.362	0.352	0.352

Notes: Standard errors clustered at the firm-country level are reported in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively. “All products” includes both products that the firm did not import in the previous year but begins importing in the current year (i.e., new products) and products that the firm continues to import from the previous year (i.e., existing products). AD contribution is calculated as the estimated coefficient of AD dummy in each specification divided by the corresponding mean probability.

6.3 Sourcing Adjustment of Non-Targeted Imports

We next investigate whether AD investigations affect firms' sourcing decisions for *non-targeted imports*. Under the search capacity constraint, disruptions induced by AD investigations may not only alter sourcing decisions for directly affected, targeted imports, but also spill over to other products imported by the same firm. We therefore aim to identify this indirect, within-firm propagation of AD shocks by exploiting firm-level exposure to AD.

For this purpose, we estimate a specification similar to (18), but replace the firm-product AD indicator with a firm-level exposure measure:

$$\text{Entry}_{fh'i,t} \text{ or } \text{Exit}_{fh'i,t} = \alpha + \beta_1 \text{AD}_{f,t-1} + \delta_{fh'} + \omega_{h'c,t} + X_{f,t-1}\Gamma + \epsilon_{fh'i,t}, \quad (20)$$

where h' denotes a non-targeted product. $\text{Entry}_{fh'i,t}$ equals one if firm f starts importing product h' from origin i in period t , and $\text{Exit}_{fh'i,t}$ equals one if firm f stops importing h' from origin i in period $t+1$, that is, if the origin is dropped after period t . $X_{f,t-1}$ denotes log firm import value, which controls for firm size.

To identify spillovers, we restrict the sample to non-targeted products only and define treatment at the firm-year level through $\text{AD}_{f,t-1}$, which equals one if firm f was exposed to at least one AD investigation in period $t-1$, and zero otherwise. The lag structure parallels the targeted-import specification: firm-level exposure is realized in $t-1$, while sourcing responses for untreated products are measured in t . As in the targeted-import specification, both entry and exit are defined around sourcing relationships that are active in year t , so the estimated spillover is interpreted as a year- t adjustment to firm-level exposure realized in $t-1$. In other words, the regressions ask whether AD exposure elsewhere in the firm's import basket in $t-1$ changes entry into and exit from sourcing relationships for non-targeted products in year t . Because the object of interest is whether a shock to one part of the import basket spills over to other, untreated products within the same firm, all directly targeted imports are excluded from the estimation sample. The coefficient on $\text{AD}_{f,t-1}$ therefore captures whether exposure elsewhere in the firm's import basket changes sourcing decisions for products that were not themselves directly investigated.

Among affected firms, targeted imports represent only 7.59% of total firm-level import value on average, with the remainder coming from non-targeted imports. AD shocks thus directly hit only a small portion of the affected firm's import basket, making it especially informative to examine whether sourcing decisions for the much larger set of non-targeted imports also respond.

The relevant comparison here differs from that in the targeted-import regressions.

There, treatment varies at the firm-product level and firm-year fixed effects can be included. Here, the object of interest is a firm-level spillover to untreated products, so treatment necessarily varies at the firm-year level. As a result, firm-year fixed effects cannot be included. Identification instead comes from comparing non-targeted imports of exposed firms with the same firm-product pairs in non-exposed years and with other firms sourcing in the same product-origin-year market, conditional on firm-product fixed effects, product-origin-year fixed effects, and firm-level controls. As in the targeted-import specification, both entry and exit are defined around sourcing relationships that are active in year t , so the estimated spillover is interpreted as a year- t adjustment to firm-level exposure realized in $t - 1$.

The key identifying assumption is that, conditional on these fixed effects and firm-level controls, the timing of firm-level AD exposure is not correlated with other unobserved firm-wide shocks that differentially affect entry or exit for non-targeted imports. This assumption is more demanding than in the targeted-import specification precisely because firm-year fixed effects cannot be included here. The estimated coefficient should therefore be interpreted as capturing spillovers of AD exposure to the firm’s broader sourcing network, rather than the direct product-specific effect of AD on the targeted products themselves.

Prior work (e.g., [Barrot and Sauvagnat \(2016\)](#); [Bernard, Moxnes and Saito \(2019\)](#)) uses natural disasters or infrastructure disruptions to study how input-specific shocks propagate through domestic sourcing networks. A key challenge in those settings is that the underlying shocks may directly affect both suppliers and customers through local channels (e.g., transportation costs) in addition to production linkages. Our setting mitigates this concern. AD investigations raise the expected tariff only for a targeted product from targeted origins, leaving the supply conditions of a firm’s other imports unchanged. This allows us to isolate how a shock to one part of the input basket reshapes sourcing decisions elsewhere within the same firm.

We expect $\beta_1 < 0$ for both outcomes, indicating that affected firms adopt a more conservative sourcing strategy for non-targeted imports. The contrast with targeted imports is informative. When a firm is forced to reconfigure sourcing for targeted inputs, search costs and capacity constraints can crowd out experimentation elsewhere: affected firms postpone adding new origins for non-targeted imports (i.e., a decrease in $n_{h',t}$). Lower entry, in turn, dampens subsequent exit because there is less scope for pruning redundant origins (i.e., a decrease in $\mu_{h',t}(n_{h',t})$). More broadly, maintaining established origins may be attractive when switching and re-entry costs are high, as emphasized by [Barrot and Sauvagnat \(2016\)](#); [Bernard, Moxnes and Saito \(2019\)](#). AD exposure can therefore tilt firms further toward preserving existing non-targeted sourcing relationships.

Finding 2. *AD exposure dampens both entry into new origins and exit from existing*

origins for non-targeted imports. These effects are also more pronounced among firms with more limited sourcing capacity, as measured by import value and the number of sourcing origins.

Table 9 reports the estimates. Columns (1)–(3) present the entry results, and columns (4)–(6) present the exit results, separately for all products, new products, and existing products. AD exposure reduces entry for non-targeted imports, with the effect concentrated among new products, while the entry response for existing products is smaller. For exit, the coefficient on AD remains negative and statistically significant across all product groups, implying that affected firms are less likely to drop established origins for non-targeted imports.

Both the estimated effects and the implied contribution of AD shocks are less pronounced for non-targeted imports than for targeted imports. Once some imported inputs are hit by AD shocks, firms reduce the entry probability of other, non-targeted imports by 0.46 percentage points, which corresponds to an 8.4% decline relative to the average entry rate (column (1)). At the same time, firms also become less likely to abandon existing sourcing origins for those products: the exit probability falls by 0.26 percentage points, equivalent to a 5.69% decline relative to the average exit rate.

Table 9: Entry and Exit Decisions for Non-Targeted Imports

	Entry _{fh'i,t}			Exit _{fh'i,t}		
	(1) All	(2) New	(3) Existing	(4) All	(5) New	(6) Existing
AD _{f,t-1}	-0.00462*** (0.000553)	-0.00369*** (0.00136)	-0.00170*** (0.000590)	-0.00262*** (0.000484)	-0.00564*** (0.00103)	-0.00139** (0.000592)
Capacity Control	YES	YES	YES	YES	YES	YES
Firm-Product FE	YES	YES	YES	YES	YES	YES
Product-Origin-Year FE	YES	YES	YES	YES	YES	YES
Mean Prob.	5.43%	8.16%	2.98%	4.57%	5.13%	4.03%
AD Contribution	-8.47%	-4.53%	-5.70%	-5.69%	-10.92%	-3.47%
Obs.	2,158,545	529,221	1,363,668	2,001,819	529,221	1,363,668
Adj. R ²	0.221	0.332	0.175	0.277	0.289	0.274

Notes: Standard errors clustered at the firm–country level are reported in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively. “All products” includes both products that the firm did not import in the previous year but begins importing in the current year (i.e., new products) and products that the firm continues to import from the previous year (i.e., existing products).

Finally, we test whether the non-targeted response varies with firm characteristics. We augment (20) with capacity controls and their interaction with AD exposure:

$$\begin{aligned}
\text{Entry}_{fh'i,t} \text{ or } \text{Exit}_{fh'i,t} = & \beta_1 \text{AD}_{f,t-1} + \beta_2 (\text{AD}_{f,t-1} \times \text{Capacity}_{f,t-1}) + \beta_3 \text{Capacity}_{f,t-1} \\
& + \delta_{fh'} + \omega_{h'c,t} + \epsilon_{fh'i,t}.
\end{aligned}
\tag{21}$$

Table 10 reports the results using baseline measures of firm sourcing capacity. Across specifications, the interaction terms are generally positive, indicating that firms with greater sourcing capacity respond less strongly to AD shocks when adjusting sourcing decisions for non-targeted imports. In other words, although AD exposure reduces both entry into new origins and exit from existing origins for non-targeted imports on average, these dampening effects are weaker for firms with larger import values, broader sourcing networks, and thus stronger overall sourcing capacity. This pattern suggests that firms with greater capacity are better able to absorb the reallocation pressure created by AD shocks and therefore face less disruption in the rest of their sourcing portfolio. By contrast, firms with more limited sourcing capacity appear more constrained, and thus scale back adjustments for non-targeted imports more sharply when targeted imports are hit. Finding 2 summarizes these results.

Table 10: Heterogeneous Responses of Non-Targeted Imports

Panel A: Entry Decisions								
	All Products			New Products			Existing Products	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$AD_{f,t}$	-0.0457*** (0.00364)	-0.0103*** (0.000910)	0.0166 (0.0108)	-0.0277*** (0.00397)	-0.00465*** (0.00106)	0.00574 (0.0132)	-0.0630*** (0.00873)	-0.0185*** (0.00197)
$\times \log \text{Import Value}_{f,t-1}$	0.00255*** (0.000210)			0.00160*** (0.000224)			0.00358*** (0.000527)	
$\times \log \text{Number of Origins}_{f,t-1}$		0.00423*** (0.000513)			0.00300*** (0.000588)			0.00745*** (0.00110)
$\times \log \text{Average Distance of Origins}_{f,t-1}$			-0.00275** (0.00124)			-0.000864 (0.00151)		-0.000862 (0.00232)
Capacity Control	YES	YES	YES	YES	YES	YES	YES	YES
Firm-Product FE	YES	YES	YES	YES	YES	YES	YES	YES
Product-Origin-Year FE	YES	YES	YES	YES	YES	YES	YES	YES
Obs.	2,057,243	2,057,243	2,057,240	1,316,593	1,316,593	1,316,593	485,556	485,556
Adj. R ²	0.231	0.236	0.220	0.177	0.182	0.175	0.325	0.327
Panel B: Exit Decisions								
	All Products			New Products			Existing Products	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$AD_{f,t}$	-0.00682** (0.00309)	-0.00413*** (0.000783)	-0.00824 (0.00777)	0.00340 (0.00388)	-0.000301 (0.00100)	-0.00249 (0.0101)	-0.0217*** (0.00695)	-0.00655*** (0.00159)
$\times \log \text{Import Value}_{f,t-1}$	0.000296* (0.000178)			-0.000248 (0.000220)			0.00101** (0.000417)	
$\times \log \text{Number of Origins}_{f,t-1}$		0.000216 (0.000415)			-0.000867* (0.000522)			-0.0001 (0.000877)
$\times \log \text{Average Distance of Origins}_{f,t-1}$			0.000646 (0.000901)			0.000128 (0.00117)		-0.00215 (0.00183)
Capacity Control	YES	YES	YES	YES	YES	YES	YES	YES
Firm-Product FE	YES	YES	YES	YES	YES	YES	YES	YES
Product-Origin-Year FE	YES	YES	YES	YES	YES	YES	YES	YES
Obs.	1,900,542	1,900,542	1,900,539	1,316,593	1,316,593	1,316,593	485,556	485,556
Adj. R ²	0.280	0.280	0.280	0.277	0.276	0.276	0.299	0.298

Notes: Standard errors clustered at the firm-country level are reported in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively. “All products” includes both products that the firm did not import in the previous year but begins importing in the current year (i.e., new products) and products that the firm continues to import from the previous year (i.e., existing products). Capacity control incorporates the corresponding capacity measure in each specification.

Our empirical results reveal a sharp contrast between firms’ responses for targeted and non-targeted imports. For targeted imports, AD exposure increases both entry into new sourcing origins and exit from existing ones, implying greater origin churning as firms actively reconfigure disrupted supplier relationships. By contrast, for non-targeted imports, AD exposure reduces both entry and exit, indicating that firms become less active in adjusting the rest of their sourcing network once attention is diverted toward targeted products. These patterns are consistent with a theory in which firms face sourcing-capacity constraints. When AD shocks force firms to reallocate resources to reoptimize targeted imports, they intensify adjustments where disruptions are most immediate while scaling back reallocation elsewhere. Additional support for this interpretation comes from the substantial heterogeneity across firms: larger firms and those with broader sourcing networks respond less strongly to AD shocks, suggesting that greater sourcing capacity enables firms to absorb supply chain disruptions more effectively.

6.4 Robustness Checks

We assess the robustness of our empirical findings in two ways: (i) by replacing the continuous capacity measures with dummy indicators, and (ii) by exploring firm heterogeneity via firm size and productivity. Appendix A.4 reports the robustness checks.

Dummies for sourcing capacity. As a robustness check, we replace the continuous measures of sourcing capacity with dummy indicators. In each year, a firm is classified as large if it falls in the top tercile of the distribution of a given capacity variable, that is, above the 66th percentile. Reassuringly, the results remain stable under this alternative specification. This confirms that our evidence on heterogeneous responses to AD shocks is not driven by the functional-form assumption embedded in the continuous measures. The results are also robust to alternative cutoffs, such as the 75th or 80th percentile. By contrast, a lower threshold, such as the median, would classify as “large” a broad set of firms that together account for the majority of aggregate imports, which would compress meaningful heterogeneity across firms.

Heterogeneity via Firm Size and Productivity. As an additional robustness check, we examine whether the heterogeneous responses documented in the baseline can be explained more generally by firm size or productivity, rather than by sourcing capacity per se. To do so, we replace our baseline capacity measures with standard proxies for firm size (e.g., sales income and labor employment) and productivity. The results show little systematic heterogeneity along these dimensions, especially for targeted imports. But for non-targeted imports, we do find evidence that larger, more productive firms are less sensitive to AD shocks. This contrast is informative. It suggests that the heterogeneity

in firms’ responses to AD shocks is not driven simply by being larger or more productive in a broad sense, but instead by features more closely tied to firms’ import sourcing operations. In other words, the baseline measures capture a distinct dimension of sourcing capacity—firms’ ability to manage, maintain, and adjust foreign supplier relationships—that is especially relevant for understanding sourcing adjustments after AD shocks. This finding reinforces our interpretation that import performance and the breadth of sourcing networks, rather than overall firm scale or productivity alone, are the key margins shaping firms’ responses to disruptions in supplier bases.

6.5 Linking the Identification Strategies to the Theory

Our empirical strategies for targeted and non-targeted imports are intentionally asymmetric because they are designed to identify different margins of adjustment. A common empirical design would either confound the direct effect of AD on targeted products with spillovers to non-targeted products, or absorb the variation needed to identify those spillovers. For targeted imports, treatment is defined at the firm-product level and the estimation sample includes both targeted and non-targeted imports. For non-targeted imports, treatment is defined at the firm level and the estimation sample is restricted to non-targeted imports only. Moreover, firm-year fixed effects can be included in the former specification but not in the latter.

The targeted-import specification is designed to identify the *direct*, product-specific response to AD. Since AD investigations are product-origin specific, the relevant question is whether a firm reconfigures its supplier base for the product directly hit by the shock. This makes the firm-product the natural level at which treatment should be defined. With firm-year fixed effects, the regressions net out any firm-wide response common across all products in a given year, so the estimated coefficient captures the differential adjustment of directly exposed imports relative to the firm’s other products.

An alternative strategy for targeted imports would be to restrict the sample to previously investigated products and compare firms directly exposed to AD with firms importing the same products only from non-targeted origins. Although this approach mirrors the descriptive comparison in Table 3, we do not pursue it for two reasons. First, the sample of targeted imports is already limited, so such a restriction would substantially reduce variation and weaken statistical power. More importantly, the direct effect of AD on targeted imports is inherently product-specific: the shock applies to particular firm-product sourcing links rather than uniformly across the firm. A firm-level comparison within the subsample of investigated products would therefore be less closely aligned with the estimand of interest and would discard the within-firm variation that is central to our preferred design.

By contrast, the non-targeted-import specification is designed to identify an *indirect*, firm-specific effect: whether a shock to one part of the import basket propagates to other products within the same firm whose own sourcing relationships are not directly targeted. This is inherently a firm-level question, so treatment must be defined at the firm level. At the same time, all targeted imports must be excluded from the estimation sample; otherwise, the estimated effect would mix the direct response of shocked products with the indirect response of untreated products. Because treatment varies at the firm-year level in this exercise, firm-year fixed effects cannot be included. The resulting specification therefore relies on observed firm-level controls and should be interpreted as evidence on within-firm spillovers, rather than the same tightly identified product-specific response recovered for targeted imports.

The asymmetry in identification aligns closely with the mechanism of our model. The model does not predict a uniform firm-wide contraction or expansion of sourcing after an AD shock. Rather, it predicts a reallocation of scarce search and supplier-management capacity across products. When a product is hit by AD, the firm has stronger incentives to search for replacement origins for that product, generating greater churning in the supplier network of targeted imports. At the same time, because search capacity is limited, resources are reallocated away from the rest of the import basket, making firms less active in adjusting non-targeted imports.

Seen in this light, the two specifications are best viewed as complements. What distinguishes our mechanism is the coexistence of *more* churning where the shock directly hits and *less* churning elsewhere within the same firm. More broadly, the two empirical strategies help delineate the scope of our central findings. Each is designed to identify a distinct margin of adjustment, and together they trace how a narrowly targeted policy shock translates into a broader firm-level adjustment mechanism. This is precisely the link between theory and stylized facts that motivates our analysis. Understanding how protectionism reshapes global sourcing networks therefore requires identifying both the direct and indirect margins of adjustment.

7 Counterfactual Analysis

What would aggregate imports have looked like in the absence of China’s AD investigations? More broadly, how much has AD policy contributed to China’s aggregate imports? We address these questions by combining our reduced-form estimates of AD-induced changes in origin entry and exit with a simple back-of-the-envelope calculation.

The exercise focuses on the extensive margin. Using the estimated effects on entry and exit, we construct the implied change in import values while holding the intensive

margin of continuing relationships fixed. This assumption is not intended to suggest that intensive-margin responses are absent; rather, it allows us to isolate the contribution of origin entry and exit, the margin directly identified by our baseline regressions. Because the exercise abstracts from changes in import values within continuing origin-product relationships, the estimates should be interpreted as the extensive-margin component of the third-country response.

Table 7 shows that for targeted imports, AD investigations induce additional entry into third-country origins but also trigger additional exit soon afterward. The counterfactual effect on import value therefore depends on the relative strength of these two forces. Aggregate effects operate through both the extensive margin, namely the number of entry and exit events, and the intensive margin, namely the value associated with each event. Our reduced-form estimates identify the extensive-margin effects. To map these into import values, we assume that AD-induced additional entry and exit carry the average import value of entry and exit observed in the same year. We also allow for dynamic effects: additional entry generates continued imports in subsequent periods, whereas additional exit eliminates future import opportunities from the exited origins. In the counterfactual without AD investigations over 2001–2015, entry for targeted imports would have been lower, reducing targeted-import value, while exit would also have been lower, raising targeted-import value.

Specifically, for each year t , we calculate the number of newly created import transactions for targeted imports (firm-product-country triples), $n_{entry,t}^{targeted}$, together with their average import value, $v_{entry,t}^{targeted}$, among affected firms. We then attribute a share $x_{entry,t}^{targeted} / (1 + x_{entry,t}^{targeted})$ of these new transactions to the AD shock in the previous year. This gives the number of AD-induced new entries:

$$\tilde{n}_{entry,t}^{targeted} = n_{entry,t}^{targeted} \times \frac{x_{entry,t}^{targeted}}{1 + x_{entry,t}^{targeted}}.$$

The contemporaneous import value generated by these AD-induced entries is assumed to equal the number of such entries multiplied by the average value per new entry:

$$\tilde{v}_{entry,t}^{targeted} = \tilde{n}_{entry,t}^{targeted} \times v_{entry,t}^{targeted}.$$

To capture persistence, we further assume that new entries in year t become incumbent import transactions in subsequent years. Their contribution to import flows in year $t + i$ is therefore computed as the number of AD-induced entries multiplied by the average import value of continuing, non-newly-added targeted transactions in that year:

$$\tilde{n}_{entry,t}^{targeted} \times v_{no-entry,t+i}^{targeted}.$$

Using this procedure, we estimate the additional import flows generated by AD-induced entry for targeted imports. We apply an analogous calculation to exit decisions to recover the import value lost because of AD-induced exit. Under these assumptions, our estimated effects of AD shocks should be interpreted as idealized upper-bound estimates.

Figure 4, Panel A, decomposes the counterfactual change in targeted-import value into two components: the decline associated with fewer entries (Lost Entry, solid line) and the increase associated with fewer exits (Lost Exit, dashed line). In the absence of AD investigations, fewer entries would have reduced China’s targeted imports by about 5%, while fewer exits would have raised them by a roughly similar magnitude. These counterfactual effects are especially pronounced during the 2008 financial crisis, mainly because the average import value of surviving targeted import transactions was substantially higher in that period than in other years.

Panel B reports the net effect. The net counterfactual is positive, indicating that the effect of fewer exits dominates that of fewer entries. In other words, because AD investigations induce firms to drop more sourcing origins than they add, the absence of AD policy would have increased China’s imports of targeted products overall. Although firms would have had weaker incentives to diversify into new origins without AD investigations, the reduction in origin exit more than offsets this effect.

AD investigations also affect non-targeted imports. From Table 9, we find that affected firms are less likely to enter new origins and less likely to drop existing origins for non-targeted products, consistent with a search-capacity constraint: firms reallocate attention toward reconfiguring targeted inputs, crowding out adjustment elsewhere. In the counterfactual without AD investigations, affected firms would instead exhibit more entry and more exit for non-targeted imports. As with targeted imports, dynamic effects amplify these changes over time.

We apply the same logic to non-targeted imports. For each year t , we compute the number of newly created non-targeted import transactions, $n_{entry,t}^{non-targeted}$, and their average import value, $v_{entry,t}^{non-targeted}$, among affected firms. We then assume that a share $x_{entry,t}^{non-targeted}/(1 - x_{entry,t}^{non-targeted})$ of these transactions would have occurred in the absence of AD shocks in the previous year, yielding

$$\tilde{n}_{entry,t}^{non-targeted} = n_{entry,t}^{non-targeted} \times \frac{x_{entry,t}^{non-targeted}}{1 - x_{entry,t}^{non-targeted}}.$$

The contemporaneous import value associated with these missing entries is

$$\tilde{v}_{entry,t}^{non-targeted} = \tilde{n}_{entry,t}^{non-targeted} \times v_{entry,t}^{non-targeted}.$$

To allow for persistence, we assume that these missing entries in year t would have

become incumbent import transactions in subsequent years. Their contribution to import flows in year $t + i$ is therefore given by

$$\tilde{n}_{entry,t}^{non-targeted} \times v_{non-entry,t+i}^{non-targeted}$$

where $v_{non-entry,t+i}^{non-targeted}$ denotes the average import value of continuing, non-newly-added non-targeted transactions in year $t + i$. Using this procedure, we recover the import flows that would have been generated by entry of non-targeted imports in the counterfactual without AD investigations. We apply the same approach to exit decisions to estimate the import value lost because of additional exit. As in the targeted case, these estimates should be interpreted as idealized upper bounds.

Figure 4, Panel C, shows the counterfactual change in non-targeted import value driven by greater entry (Additional Entry, solid line) and the offsetting decline driven by greater exit (Additional Exit, dashed line). These changes are measured relative to the total value of non-targeted imports purchased by *affected firms*, since we assume that non-affected firms do not adjust their supplier bases. Relative to total non-targeted imports by all firms, the implied effects would therefore be even smaller.

In the absence of AD investigations, more entry would have increased non-targeted imports by about 0.5%, while more exit would have reduced them by a broadly similar magnitude. Since AD shocks crowd out sourcing adjustments for non-targeted products, affected firms become less active on both the entry and exit margins outside the targeted products. Without such disruption, firms actively adjust their supplier bases with additional entry and exit.

Panel D reports the net counterfactual effect on non-targeted imports. The effect is positive throughout, indicating that, in the absence of AD investigations, non-targeted imports by affected firms would have been higher in every year. This positive net effect implies that the gain from additional entry into new origins outweighs the loss from additional exit from existing origins. Put differently, although firms would have dropped some origins more frequently without AD shocks, the expansion along the entry margin would have been stronger overall.

Combining the results, the counterfactual implies that China's aggregate imports would have been higher in the absence of AD investigations. This is driven by both targeted and non-targeted imports. AD investigations reshape import flows for affected firms by altering origin churning across all imports. For targeted imports, removing AD shocks would make firms less likely both to search for new origins and to abandon existing ones. In net terms, the stabilizing effect from lower exit dominates, leading to higher import values for targeted products. For non-targeted imports, by contrast, the absence of AD investigations would make affected firms more active along both the entry

and exit margins, with the positive effect of additional entry dominating overall.

Taken together, these findings suggest that the trade effects of AD policy extend well beyond the directly investigated products from targeted origins. Even for targeted imports, the net effect depends on the competing effects on entry and exit. More importantly, focusing only on targeted imports would miss a key channel through which AD reduces trade: its dampening effect on sourcing adjustments for non-targeted imports within the same firm. Ignoring this spillover would therefore understate the overall trade-reducing effect of AD policy on China's imports.

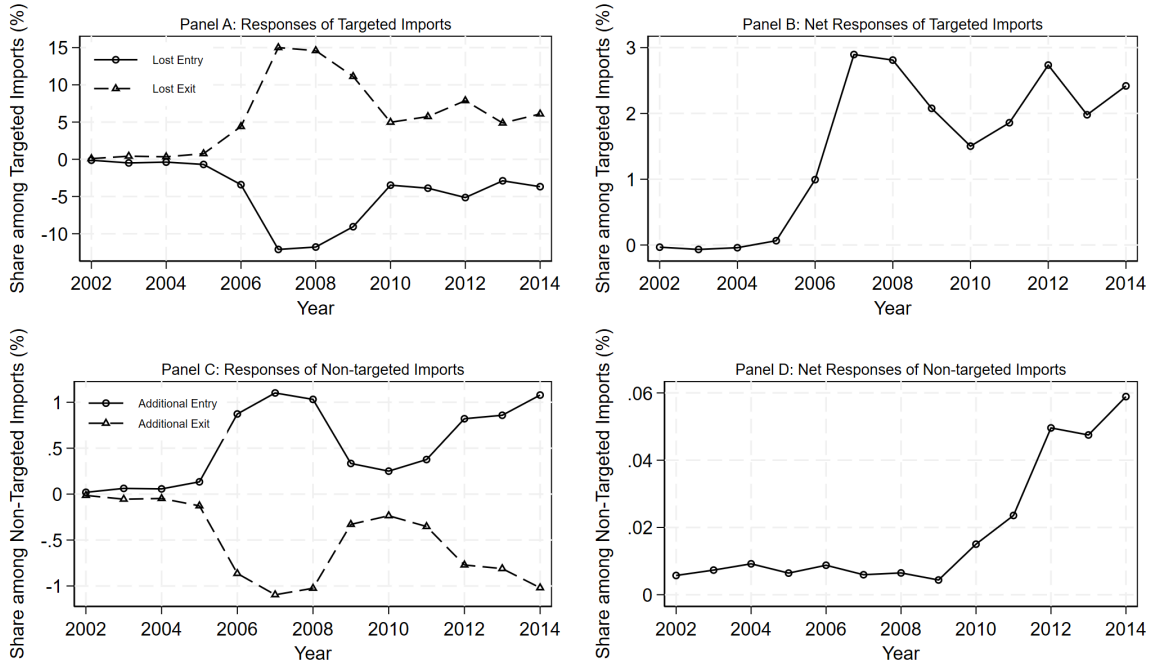


Figure 4: Counterfactual Changes in Aggregate Import Value Without Antidumping

Notes: The figure presents back-of-the-envelope estimates of changes in aggregate import values for targeted and non-targeted imports under a counterfactual scenario without AD shocks. The percentage changes are reported relative to the observed total import values of targeted and non-targeted imports for affected firms. The exercise combines reduced-form estimates of AD-induced changes in entry and exit with observed average import values by year and transaction type, while allowing these effects to persist over time. For targeted imports, Panel A shows the effects of fewer entries and fewer exits in the absence of AD investigations. For non-targeted imports, Panel C shows the effects of additional entry and additional exit in the absence of AD investigations. Panels B and D report the corresponding net changes.

8 Conclusions

China's antidumping activity expanded sharply after WTO accession, raising the expected cost of importing specific product–origin pairs and generating recurrent disruptions to firms' foreign sourcing relationships. Such policy-driven shocks force importers

not only to adjust the directly affected supplier links, but also to reallocate procurement effort across the rest of their input basket.

This paper studies how firms reconfigure foreign supplier bases in response to these disruptions. We develop a parsimonious dynamic model in which expanding and maintaining a supplier base requires scarce sourcing capacity, so firms cannot simultaneously reoptimize all sourcing relationships. Guided by the model, we exploit China’s AD investigations to identify shocks to narrowly defined product–origin relationships and estimate entry and exit responses along the extensive margin.

Our results show sharp and asymmetric adjustments across targeted and non-targeted imports. For targeted imports, AD exposure increases both origin entry and origin exit: affected firms actively search for alternative third-country origins, but they also shed established origins shortly thereafter, consistent with rapid reconfiguration and within-product trade diversion. These responses are heterogeneous: firms with broader supplier networks and greater sourcing capacity adjust less along both margins, consistent with a stronger ability to absorb disruptions within their existing networks. For non-targeted imports, in contrast, AD exposure reduces both entry into new origins and exit from existing origins, with limited evidence of size heterogeneity. Taken together, these patterns indicate that sourcing decisions are not separable across inputs in the short run. When firms are forced to reconfigure sourcing for targeted inputs, capacity constraints crowd out adjustment elsewhere, dampening origin churning for non-targeted inputs.

Our findings contribute to the literature on trade policy and firm adjustment by documenting within-firm propagation of policy-induced supply-chain disruptions and by highlighting a simple mechanism—limited sourcing capacity—that links targeted shocks to broader changes in procurement dynamics. They also suggest that evaluating the effects of trade shocks requires tracking firm-wide sourcing responses, not only the directly targeted relationships.

Several extensions are natural. One direction is to quantify longer-run adjustment by tracing dynamics beyond the one-year horizon and examining persistence in newly formed and severed links. A second is to discipline mechanisms using a structural framework that connects observed churning to costs of search, switching, and re-entry. A third is to assess external validity by applying the same empirical design to other sources of input-specific disruptions, such as tariffs, sanctions, or logistics shocks, to study how different shock characteristics map into supplier-base reconfiguration.

References

- Acemoglu, Daron, Vasco M Carvalho, Asuman Ozdaglar, and Alireza Tahbaz-Salehi.** 2012. “The network origins of aggregate fluctuations.” *Econometrica*, 80(5): 1977–2016.
- Akerberg, Daniel A, Kevin Caves, and Garth Frazer.** 2015. “Identification properties of recent production function estimators.” *Econometrica*, 83(6): 2411–2451.
- Ahn, JaeBin, Amit K Khandelwal, and Shang-Jin Wei.** 2011. “The role of intermediaries in facilitating trade.” *Journal of International Economics*, 84(1): 73–85.
- Amiti, Mary, Oleg Itskhoki, and Jozef Konings.** 2014. “Importers, exporters, and exchange rate disconnect.” *American Economic Review*, 104(7): 1942–78.
- Antras, Pol, Teresa C Fort, and Felix Tintelnot.** 2017. “The margins of global sourcing: Theory and evidence from U.S. firms.” *American Economic Review*, 107(9): 2514–64.
- Bao, Xiaohua, Bruce A. Blonigen, and Zhi Yu.** 2026. “Cross-product and cross-market adjustments within multiproduct firms: Evidence from antidumping actions.” *Journal of International Economics*, 159: 104206.
- Barrot, Jean-Noël, and Julien Sauvagnat.** 2016. “Input specificity and the propagation of idiosyncratic shocks in production networks.” *The Quarterly Journal of Economics*, 131(3): 1543–1592.
- Bernard, Andrew B, Andreas Moxnes, and Yukiko U Saito.** 2019. “Production networks, geography, and firm performance.” *Journal of Political Economy*, 127(2): 639–688.
- Bernard, Andrew B, J Bradford Jensen, Stephen J Redding, and Peter K Schott.** 2018. “Global firms.” *Journal of Economic Literature*, 56(2): 565–619.
- Blaum, Joaquin.** 2024. “Global firms in large devaluations.” *The Quarterly Journal of Economics*, 139(4): 2427–2474.
- Blaum, Joaquin, Claire Lelarge, and Michael Peters.** 2019. “Firm size, quality bias and import demand.” *Journal of International Economics*, 120: 59–83.
- Bown, Chad P.** 2016. “Global antidumping database.” *World Bank*.
- Bown, Chad P, and Meredith A Crowley.** 2006. “Policy externalities: How US antidumping affects Japanese exports to the EU.” *European Journal of Political Economy*, 22(3): 696–714.

- Bown, Chad P, and Meredith A Crowley.** 2007. "Trade deflection and trade depression." *Journal of International Economics*, 72(1): 176–201.
- Bown, Chad P, Paola Conconi, Aksel Erbahar, and Lorenzo Trimarchi.** 2021. "Trade protection along supply chains."
- Brandt, Loren, Johannes Van Biesebroeck, and Yifan Zhang.** 2012. "Creative accounting or creative destruction? Firm-level productivity growth in Chinese manufacturing." *Journal of Development Economics*, 97(2): 339–351.
- Brenton, Paul.** 2001. "Anti-dumping policies in the EU and trade diversion." *European Journal of Political Economy*, 17(3): 593–607.
- Caliendo, Lorenzo, and Fernando Parro.** 2015. "Estimates of the Trade and Welfare Effects of NAFTA." *The Review of Economic Studies*, 82(1): 1–44.
- Carvalho, Vasco M, Makoto Nirei, Yukiko Saito, and Alireza Tahbaz-Salehi.** 2016. "Supply chain disruptions: Evidence from the great east japan earthquake." *Columbia Business School Research Paper*, , (17-5).
- Chandra, Piyush, and Cheryl Long.** 2013. "Anti-dumping duties and their impact on exporters: Firm level evidence from China." *World development*, 51: 169–186.
- Chen, Ming, Wen Chen, Ling Feng, Zhiyuan Li, and Deborah L. Swenson.** 2026. "The chilling effects of unsuccessful antidumping investigations on firm export dynamics." *Journal of Development Economics*, 178: 103610.
- Crowley, Meredith, Ning Meng, and Huasheng Song.** 2018. "Tariff scares: Trade policy uncertainty and foreign market entry by Chinese firms." *Journal of International Economics*, 114: 96–115.
- Eaton, Jonathan, and Samuel Kortum.** 2002. "Technology, geography, and trade." *Econometrica*, 70(5): 1741–1779.
- Ganguli, Bodhisattva.** 2008. "The trade effects of Indian antidumping actions." *Review of International Economics*, 16(5): 930–941.
- Gopinath, Gita, and Brent Neiman.** 2014. "Trade adjustment and productivity in large crises." *American Economic Review*, 104(3): 793–831.
- Grossman, Gene M, Elhanan Helpman, and Stephen J Redding.** 2024. "When tariffs disrupt global supply chains." *American Economic Review*, 114(4): 988–1029.
- Jabbour, Liza, Zhigang Tao, Enrico Vanino, and Yan Zhang.** 2019. "The good, the bad and the ugly: Chinese imports, European Union anti-dumping measures and firm performance." *Journal of International Economics*, 117: 1–20.

- Konings, Jozef, and Hylke Vandenbussche.** 2005. “Antidumping protection and markups of domestic firms.” *Journal of International Economics*, 65(1): 151–165.
- Konings, Jozef, and Hylke Vandenbussche.** 2008. “Heterogeneous responses of firms to trade protection.” *Journal of International Economics*, 76(2): 371–383.
- Krupp, Corinne M, and Patricia S Pollard.** 1996. “Market responses to antidumping laws: Some evidence from the US chemical industry.” *Canadian Journal of Economics*, 199–227.
- Lu, Dan, Asier Mariscal, and Luis-Fernando Mejía.** 2024. “How firms accumulate inputs: Evidence from import switching.” *Journal of International Economics*, 148: 103847.
- Lu, Yi, Zhigang Tao, and Yan Zhang.** 2013. “How do exporters respond to antidumping investigations?” *Journal of International Economics*, 91(2): 290–300.
- Lu, Yi, Zhigang Tao, and Yan Zhang.** 2018. “How do exporters adjust export product scope and product mix to react to antidumping?” *China Economic Review*, 51: 20–41.
- Messerlin, Patrick A.** 2004. “China in the World Trade Organization: Antidumping and Safeguards.” *World Bank Economic Review*, 18(1): 105–130.
- Olley, G Steven, and Ariel Pakes.** 1996. “The Dynamics of Productivity in the Telecommunications Equipment Industry.” *Econometrica*, 64(6): 1263–1297.
- Pankratz, Nora MC, and Christoph M Schiller.** 2024. “Climate change and adaptation in global supply-chain networks.” *The Review of Financial Studies*, 37(6): 1729–1777.
- Park, Seung Ho, and Gerardo R Ungson.** 2001. “Interfirm rivalry and managerial complexity: A conceptual framework of alliance failure.” *Organization science*, 12(1): 37–53.
- Park, Soonchan.** 2009. “The trade depressing and trade diversion effects of antidumping actions: The case of China.” *China Economic Review*, 20(3): 542–548.
- Petrin, Amil, and James Levinsohn.** 2012. “Measuring aggregate productivity growth using plant-level data.” *The RAND Journal of Economics*, 43(4): 705–725.
- Pierce, Justin R.** 2011. “Plant-level responses to antidumping duties: Evidence from US manufacturers.” *Journal of International Economics*, 85(2): 222–233.
- Prusa, Thomas J.** 1997. “The trade effects of US antidumping actions.” In *The effects of US trade protection and promotion policies*. 191–214. University of Chicago Press.

- Prusa, Thomas J.** 2001. “On the spread and impact of anti-dumping.” *Canadian Journal of Economics/Revue canadienne d’économique*, 34(3): 591–611.
- Rozemeijer, Frank A, Arjan Van Weele, and Mathieu Weggeman.** 2003. “Creating corporate advantage through purchasing: toward a contingency model.” *Journal of Supply Chain Management*, 39(4): 4–13.
- Staiger, Robert W, and Frank A Wolak.** 1994. “Measuring industry specific protection: Antidumping in the United States.”
- Van, LTT, and SY Tong.** 2009. “China and Anti-Dumping: Regulations, Practices and Responses.” *East Asia Institute. National University of Singapore Working Paper*, , (149).
- Wang, Lei, and Shengxing Yu.** 2002. “China’s New Anti-dumping Regulations Improvements to Comply with the World Trade Organization Rules.” *Journal of World Trade*, 36(5).
- Zhou, Eric Wentao, Kim J. Ruhl, and Xing Xu.** 2026. “The Aggregate Impact of Anti-dumping Policy: Theory and Evidence.” Unpublished manuscript (work in progress).

A Empirical Appendix

This empirical appendix provides additional details on summary statistics and robustness checks.

A.1 Constructions of Key Variables

In the baseline analysis, we focus on *third-country* responses for targeted products, that is, adjustments in origins other than the origin(s) directly targeted in $t - 1$. We implement this design by dropping two sets of observations. First, if (h, j) is targeted in $t - 1$ and firm f still imports (h, j) in t , we drop (f, h, j) in period t so that entry/exit is evaluated only in non-targeted (third-country) origins.¹⁹ Second, if (h', i) is targeted in period t , we drop (f, h', i) in t to remove contemporaneous effects and identify the one-period lagged response.

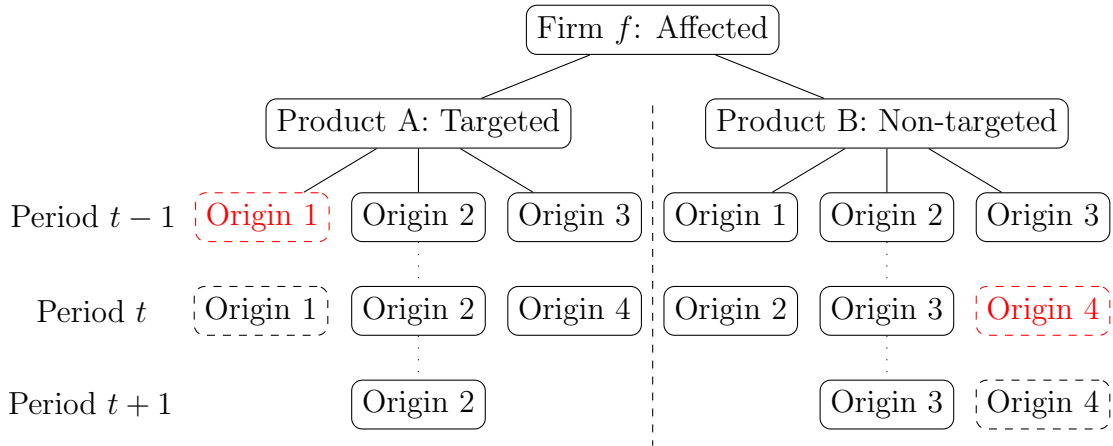


Figure 5: Import Sourcing Pattern of Firm: An Example

Notes: Dashed boxes indicate observations that are excluded from the relevant estimating sample. Red dashed boxes identify directly targeted origins for targeted imports.

Figure 5 provides an example illustrating firm f 's sourcing profile over $t - 1$, t , and $t + 1$. The firm imports two products, A and B , in all three periods but changes origins over time: in $t - 1$ it sources both products from origins 1, 2, 3, while in t it sources A from 1, 2, 4 and B from 2, 3, 4. Product A from origin 1 is subject to an AD shock in $t - 1$ (red), whereas product B is not. Hence, A is targeted in t , B is non-targeted, and the firm is affected because it imported $(A, \text{origin 1})$ in $t - 1$.

¹⁹We do not study the direct effect of AD on imports from the targeted origin itself, that is, whether an AD investigation on product h from origin j in $t - 1$ affects sourcing from j in t . We provide evidence that AD shocks deter imports of targeted products from targeted origins, consistent with the standard trade depressing effect.

Table A1: Construction of AD, Entry and Exit Dummy at Period t : An Example

	Product A: Targeted			Product B: Non-targeted		
	Origin 1	Origin 2	Origin 4	Origin 2	Origin 3	Origin 4
$AD_{fh,t-1}$	1	1	1	0	0	0
$AD_{f,t-1}$	1	1	1	1	1	1
$Entry_{fhi,t}$	0	0	1	0	0	1
$Exit_{fhi,t}$	1	0	1	1	0	0

Table A1 reports the implied indicator values in period t . Since A is targeted, $AD_{fh,t-1} = 1$ for all (f, A, i) observations in t , whereas $AD_{fh,t-1} = 0$ for all (f, B, i) . Because the firm is affected, $AD_{f,t-1} = 1$ for all its observations in t . Entry compares t with $t - 1$: $Entry_{fhi,t} = 1$ for origin 4 of both products and 0 otherwise. Exit compares t with $t + 1$: the firm stops sourcing A from origins 1 and 4 and B from origin 2, so $Exit_{fhi,t} = 1$ for those observations and 0 otherwise. Appendix A.2 reports the summary statistics for empirical analysis of firm sourcing decisions. Given the timing and construction of the measures, we trace how a shock to a specific product–origin sourcing relationship shapes firms’ subsequent adjustments in both directly targeted imports and other, non-targeted imports within the same firm. Two additional patterns are worth highlighting. First, for targeted products, imports from the targeted origin account for 84.7% of a firm’s total import value of targeted imports, on average. This implies that AD investigations typically generate a sizable shock to targeted imports, so we should expect meaningful adjustments in their sourcing patterns. Second, among affected firms, targeted imports represent only 7.59% of total firm-level import value on average, with the remaining coming from non-targeted imports. Thus, AD shocks directly hit only a small portion of an affected firm’s import activity, making it particularly informative to examine whether sourcing decisions for non-targeted imports—which account for the bulk of imports—also respond.

A.2 Summary Statistics

Table A2 reports summary statistics for the transaction-level sample. Table A3 summarizes firm-level characteristics for the matched CCD–ASIE sample. Firm-level import value is constructed from the CCD, while the remaining variables are from the ASIE. Affected firms constitute only a small share of the sample, but Table A3 indicates that they are, on average, larger than unaffected firms (Panel C) in terms of size, productivity, and import value (Panel B).

Table A2: Summary Statistics: Transaction Sample

Panel A: All Firms						
	All		New		Existing	
	Mean	Sd	Mean	Sd	Mean	Sd
$AD_{f,ht-1}$	0.0011	0.340	0	0	0.0019	0.0433
$AD_{f,t-1}$	0.1283	0.3345	0.1040	0.3053	0.1436	0.3507
$Entry_{fhi,t}$	0.0618	0.2409	0.0923	0.2894	0.0428	0.2023
$Exit_{fhi,t}$	0.0524	0.2228	0.0521	0.2222	0.0470	0.2115
# of Firms	3,636		3,636		3,636	
# of Firm-Product Pairs	576,947		561,535		257,838	
Obs.	2,576,407		992,216		1,584,191	
Panel B: Affected Firms						
	All		New		Existing	
	Mean	Sd	Mean	Sd	Mean	Sd
AD_{fht-1}	0.0090	0.0946	-	-	0.0131	0.1139
$AD_{f,t-1}$	1	0	1	0	1	0
$Entry_{fhi,t}$	0.0308	0.1728	0.0421	0.2009	0.0257	0.1582
$Exit_{fhi,t}$	0.0323	0.1768	0.0321	0.1763	0.0315	0.1747
# of Firms	1,782		1,756		1,757	
# of Firm-Product Pairs	153,774		85,603		86,115	
Obs.	330,642		103,204		227,438	
Panel C: Unaffected Firms						
	All		New		Existing	
	Mean	Sd	Mean	Sd	Mean	Sd
AD_{fht-1}	0	0	-	-	0	0
$AD_{f,t-1}$	0	0	0	0	0	0
$Entry_{fhi,t}$	0.0664	0.2490	0.0981	0.2974	0.0456	0.2087
$Exit_{fhi,t}$	0.0555	0.2290	0.0544	0.2268	0.0495	0.2170
# of Firms	3,636		3,636		3,633	
# of Firm-Product Pairs	548,307		522,738		245,255	
Obs.	2,082,028		889,012		1,356,753	

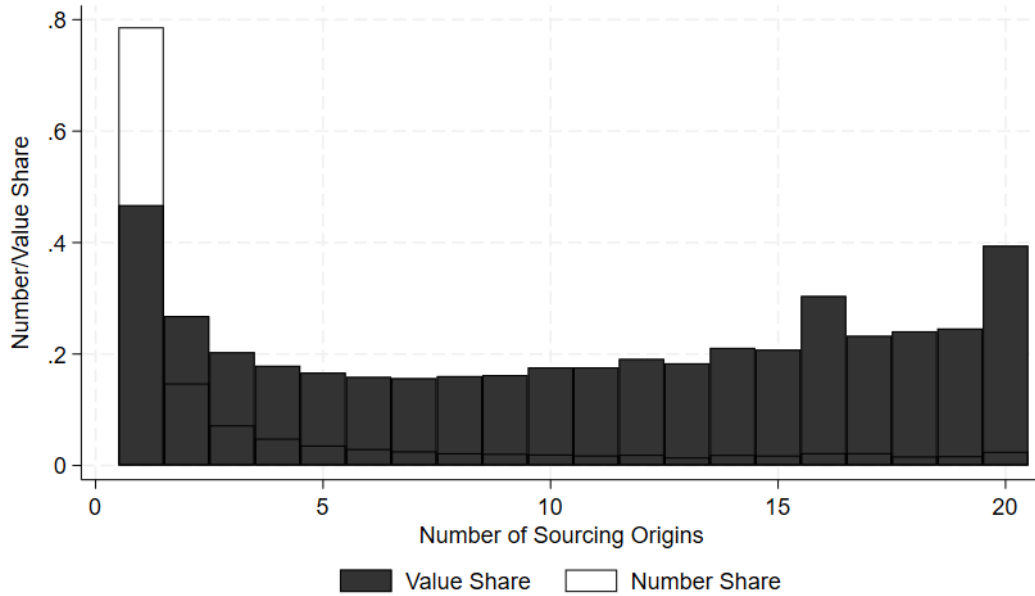
Table A3: Summary Statistics: Firm-level Characteristics

Panel A: All Firms		
	Mean	Sd
$\ln(\text{Sales})_{f,t-1}$	7.3504	1.5748
$\ln(\text{Labor})_{f,t-1}$	5.8401	1.3726
$\text{TFP}(\text{OP})_{f,t-1}$	0.8873	0.3608
$\text{TFP}(\text{LP})_{f,t-1}$	1.5278	0.4336
$\text{TFP}(\text{ACF})_{f,t-1}$	0.2988	0.5998
# of Firms	3,072	
Obs.	21,472	
	Mean	Sd
$\ln(\text{Import Value})_{f,t-1}$	14.0258	2.6699
Number of Origins $_{f,t-1}$	6.3612	5.8878
$\ln(\text{Average Distance from Origins})_{f,t-1}$	8.5756	0.5527
# of Firms	3,636	
Obs.	50,580	
Panel B: Affected Firms		
	Mean	Sd
$\ln(\text{Sales})_{f,t-1}$	8.0131	1.6361
$\ln(\text{Labor})_{f,t-1}$	6.2161	1.4279
$\text{TFP}(\text{OP})_{f,t-1}$	0.9607	0.3936
$\text{TFP}(\text{LP})_{f,t-1}$	1.6546	0.4502
$\text{TFP}(\text{ACF})_{f,t-1}$	0.3821	0.6358
# of Firms	1,128	
Obs.	1,675	
	Mean	Sd
$\ln(\text{Import Value})_{f,t-1}$	15.4777	2.1952
Number of Origins $_{f,t-1}$	10.6120	8.1454
$\ln(\text{Average Distance from Origins})_{f,t-1}$	8.6636	0.4382
# of Firms	1,783	
Obs.	3,052	
Panel C: Unaffected Firms		
	Mean	Sd
$\ln(\text{Sales})_{f,t-1}$	7.2943	1.5566
$\ln(\text{Labor})_{f,t-1}$	5.8080	1.3630
$\text{TFP}(\text{OP})_{f,t-1}$	0.8811	0.3572
$\text{TFP}(\text{LP})_{f,t-1}$	1.5172	0.4305
$\text{TFP}(\text{ACF})_{f,t-1}$	0.2919	0.5962
# of Firms	3,047	
Obs.	19,797	
	Mean	Sd
$\ln(\text{Import Value})_{f,t-1}$	13.9326	2.6707
Number of Origins $_{f,t-1}$	6.0882	5.6034
$\ln(\text{Average Distance from Origins})_{f,t-1}$	8.5699	0.5588
# of Firms	3,636	
Obs.	47,528	

A.3 Multi-Origin Imports

This section presents evidence on the prominent role of multi-origin imports (i.e., inputs sourced from more than one foreign country) in shaping both firm-level and aggregate import flows. Figure 6 provides a more detailed breakdown of how imports with varying numbers of sourcing origins contribute to firms' overall import activity.

Figure 6: Foreign Supplier Bases of Chinese Firms



Notes: This figure shows the (average) number and value share of imports with varying numbers of origins within a firm. The calculations are performed for each firm in each year, followed by averaging across firms and time periods.

A.4 Robustness Checks

This appendix reports a range of robustness checks designed to verify that our main empirical results are not driven by particular choices in measuring sourcing capacity, nor by heterogeneity in firm size or productivity.

Dummies for sourcing capacity. Tables A4–A5 report the estimation results using dummy measures of sourcing capacity. In each year, a firm is classified as large if it falls in the top tercile of the distribution of a given capacity variable. Reassuringly, the results remain broadly consistent with the baseline specifications that use continuous measures.

Table A4: Entry and Exit Decisions for Targeted Imports: Capacity Dummies

Panel A: Entry Decisions						
	All Products			New Products		
	(1)	(2)	(3)	(4)	(5)	(6)
$AD_{fh,t-1}$	0.0271** (0.0129)	0.0508*** (0.0142)	0.0131* (0.00684)	0.0209 (0.0134)	0.0556*** (0.0154)	0.0212*** (0.00676)
$\times$ Large Import Value $_{f,t-1}$	-0.0163 (0.0127)			-0.00346 (0.0134)		
$\times$ Large Number of Origins $_{f,t-1}$		-0.0437*** (0.0143)			-0.0436*** (0.0157)	
$\times$ Large Average Distance $_{f,t-1}$			0.00133 (0.00810)			-0.00697 (0.00807)
Obs.	2,158,187	2,158,187	2,158,187	1,359,959	1,359,959	1,359,959
Adj. R ²	0.360	0.360	0.360	0.226	0.226	0.226
Panel B: Exit Decisions						
	All Products			New Products		
	(1)	(2)	(3)	(4)	(5)	(6)
$AD_{fh,t-1}$	0.0240** (0.0116)	0.0307** (0.0137)	0.0180*** (0.00649)	0.0252** (0.0122)	0.0283* (0.0145)	0.0160** (0.00674)
$\times$ Large Import Value $_{f,t-1}$	-0.0106 (0.0115)			-0.0146 (0.0119)		
$\times$ Large Number of Origins $_{f,t-1}$		-0.0182 (0.0137)			-0.0178 (0.0146)	
$\times$ Large Average Distance $_{f,t-1}$			-0.00599 (0.00758)			-0.00676 (0.00773)
Obs.	2,001,753	2,001,753	2,001,753	1,359,959	1,359,959	1,359,959
Adj. R ²	0.360	0.360	0.360	0.349	0.349	0.349

Notes: Standard errors clustered at the firm–country level are reported in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively. All exercises use specification (18) and include the specified fixed effects.

Table A5: Entry and Exit Decisions for Targeted Imports: Capacity Dummies

Panel A: Entry Decisions									
	All Products			New Products			Existing Products		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
AD _{<i>f,t-1</i>}	-0.00196* (0.00110)	-0.0153*** (0.00116)	-0.00501*** (0.000733)	-0.00580*** (0.00127)	-0.00786*** (0.00142)	-0.00131 (0.000885)	-0.0146*** (0.00227)	-0.0168*** (0.00214)	-0.00776*** (0.00163)
.× Large Import Value _{<i>f,t-1</i>}	-0.00319*** (0.00111)			0.00641*** (0.00127)			0.0127*** (0.00234)		
.× Large Number of Origins _{<i>f,t-1</i>}		0.0159*** (0.00119)			0.00962*** (0.00145)			0.0162*** (0.00229)	
.× Large Average Distance _{<i>f,t-1</i>}			-0.00371*** (0.000904)			-0.000765 (0.00103)			-0.00337 (0.00220)
Capacity Control	YES	YES	YES	YES	YES	YES	YES	YES	YES
Obs.	2,158,545	2,057,243	2,057,240	1,316,593	1,316,593	1,316,593	485,556	485,556	485,554
Adj. R ²	0.221	0.242	0.220	0.177	0.184	0.175	0.325	0.331	0.318
Panel B: Exit Decisions									
	All Products			New Products			Existing Products		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
AD _{<i>f,t-1</i>}	-0.00254** (0.00101)	-0.00455*** (0.000876)	-0.00276*** (0.000652)	0.000539 (0.00134)	-0.00231** (0.00115)	-0.00126 (0.000846)	-0.00744*** (0.00190)	-0.00879*** (0.00170)	-0.00443*** (0.00127)
.× Large Import Value _{<i>f,t-1</i>}	0.000853 (0.00101)			-0.00160 (0.00133)			0.00265 (0.00197)		
.× Large Number of Origins _{<i>f,t-1</i>}		0.00255*** (0.000915)			0.00126 (0.00119)			0.00361** (0.00184)	
.× Large Average Distance _{<i>f,t-1</i>}			0.000248 (0.000794)			-0.000157 (0.000990)			-0.00364** (0.00176)
Capacity Control	YES	YES	YES	YES	YES	YES	YES	YES	YES
Obs.	1,900,542	1,900,542	1,900,539	1,316,593	1,316,593	1,316,593	485,556	485,556	485,554
Adj. R ²	0.280	0.280	0.280	0.277	0.276	0.276	0.299	0.298	0.298

Notes: Standard errors clustered at the firm–country level are reported in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively. All exercises use specification (20) and include the specified fixed effects.

Heterogeneity via Firm Size and Productivity. Tables A6–A9 report the results for targeted imports using specification (19), while Tables A10–A15 report the corresponding results for non-targeted imports using specification (21). In these exercises, we replace the baseline sourcing-capacity measures with alternative firm characteristics, including sales income, employment, TFP estimates, the weighted average distance to origin countries, import intensity, and the share of processing imports. These alternative measures deliver much weaker and less systematic heterogeneity than the baseline capacity proxies, especially for targeted imports. For non-targeted imports, we do find heterogeneous responses for firms with different size and productivity. This contrast suggests that firms’ responses to AD shocks are not primarily shaped by broad differences in firm size or productivity, but rather by dimensions more directly tied to import sourcing performance and the management of foreign supplier relationships. In this sense, our baseline measures capture the distinctive aspect of firm capacity that matters most for sourcing adjustment.

Table A6: Entry Decisions for Targeted Imports: All Products and Size Measures

	Dependent variable: Entry _{fhi,t}							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
AD _{fh,t-1}	-0.0157 (0.0378)	-0.0121 (0.0301)	-0.0183 (0.0223)	-0.0113 (0.0284)	0.00562 (0.00980)	-0.0386 (0.0752)	0.0211 (0.0129)	0.0118 (0.00728)
.× log Sales income _{f,t-1}	0.00287 (0.00377)							
.× log Employment _{f,t-1}		0.00330 (0.00386)						
.× TFP (OP) _{f,t-1}			0.0191 (0.0170)					
.× TFP (LP) _{f,t-1}				0.00987 (0.0130)				
.× TFP (ACF) _{f,t-1}					0.00310 (0.00791)			
.× log Weighted Distance _{f,t-1}						0.00600 (0.00883)		
.× Import Intensity _{f,t-1}							-0.0416 (0.0319)	
.× Processing Trade Share _{f,t-1}								0.00944 (0.0219)
Obs.	784,114	793,890	514,728	689,640	689,363	2,057,361	724,100	2,150,533
Adj. R ²	0.308	0.309	0.292	0.305	0.305	0.313	0.300	0.323

Notes: Standard errors clustered at the firm–country level are reported in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively. All exercises use specification (19) and include the specified fixed effects.

Table A7: Entry Decisions for Targeted Imports: Existing Products and Size Measures

	Dependent variable: Entry _{fhi,t}							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
AD _{fh,t-1}	-0.00976 (0.0376)	-0.0108 (0.0318)	-0.00804 (0.0224)	-0.0175 (0.0286)	0.00423 (0.00954)	0.0315 (0.0765)	0.0157 (0.0132)	0.0145** (0.00731)
.× log Sales income _{f,t-1}	0.00206 (0.00371)							
.× log Employment _{f,t-1}		0.00268 (0.00407)						
.× TFP (OP) _{f,t-1}			0.0160 (0.0173)					
.× TFP (LP) _{f,t-1}				0.0140 (0.0132)				
.× TFP (ACF) _{f,t-1}					0.00882 (0.00804)			
.× log Weighted Distance _{f,t-1}						-0.00174 (0.00900)		
.× Import Intensity _{f,t-1}							-0.0339 (0.0333)	
.× Processing Trade Share _{f,t-1}								0.0160 (0.0230)
Obs.	486,395	493,506	325,320	432,311	432,198	1,313,264	453,164	1,359,959
Adj. R ²	0.212	0.213	0.202	0.211	0.211	0.223	0.210	0.226

Notes: Standard errors clustered at the firm–country level are reported in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively. All exercises use specification (19) and include the specified fixed effects.

Table A8: Exit Decisions for Targeted Imports: All Products and Size Measures

	Dependent variable: Exit _{fhi,t}							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
AD _{fh,t-1}	0.0350 (0.0317)	0.0233 (0.0265)	-0.0160 (0.0162)	0.000909 (0.0224)	0.00544 (0.00901)	0.115* (0.0686)	0.0142 (0.0114)	0.0121 (0.00740)
.× log Sales income _{f,t-1}	-0.00299 (0.00318)							
.× log Employment _{f,t-1}		-0.00219 (0.00336)						
.× TFP (OP) _{f,t-1}			0.0136 (0.0108)					
.× TFP (LP) _{f,t-1}				0.00336 (0.00982)				
.× TFP (ACF) _{f,t-1}					0.00386 (0.00597)			
.× log Weighted Distance _{f,t-1}						-0.0116 (0.00803)		
.× Import Intensity _{f,t-1}							-0.0292 (0.0279)	
.× Processing Trade Share _{f,t-1}								0.0142 (0.0207)
Obs.	784,114	793,890	514,728	689,640	689,363	1,900,971	724,100	1,994,084
Adj. R ²	0.372	0.371	0.375	0.378	0.378	0.362	0.374	0.360

Notes: Standard errors clustered at the firm–country level are reported in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively. All exercises use specification (19) and include the specified fixed effects.

Table A9: Exit Decisions for Targeted Imports: Existing Products and Size Measures

	Dependent variable: Exit _{fhi,t}							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
AD _{fhi,t-1}	0.00721 (0.0337)	9.60e-05 (0.0283)	-0.0277 (0.0181)	-0.0204 (0.0253)	-0.00251 (0.00987)	0.0802 (0.0710)	0.00115 (0.0121)	0.0123 (0.00763)
.× log Sales income _{f,t-1}	-0.000869 (0.00336)							
.× log Employment _{f,t-1}		-6.12e-05 (0.00355)						
.× TFP (OP) _{f,t-1}			0.0158 (0.0120)					
.× TFP (LP) _{f,t-1}				0.0102 (0.0111)				
.× TFP (ACF) _{f,t-1}					0.00315 (0.00610)			
.× log Weighted Distance _{f,t-1}						-0.00780 (0.00832)		
.× Import Intensity _{f,t-1}							-0.0162 (0.0279)	
.× Processing Trade Share _{f,t-1}								0.00295 (0.0213)
Obs.	486,395	493,506	325,320	432,311	432,198	1,313,264	453,164	1,359,959
Adj. R ²	0.367	0.366	0.371	0.372	0.372	0.352	0.370	0.349

Notes: Standard errors clustered at the firm–country level are reported in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively. All exercises use specification (19) and include the specified fixed effects.

Table A10: Entry Decisions for Non-Targeted Imports: All Products and Size Measures

	Dependent variable: Entry _{fhi,t}							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
AD _{f,t-1}	-0.0234*** (0.00319)	-0.0172*** (0.00308)	-0.0117*** (0.00180)	-0.0168*** (0.00241)	-0.00262*** (0.000962)	-0.00967 (0.00643)	-0.00733*** (0.00106)	-0.000987 (0.000693)
.× log Sales income _{f,t-1}	0.00250*** (0.000335)							
.× log Employment _{f,t-1}		0.00240*** (0.000413)						
.× TFP (OP) _{f,t-1}			0.0110*** (0.00139)					
.× TFP (LP) _{f,t-1}				0.00871*** (0.00118)				
.× TFP (ACF) _{f,t-1}					0.00374*** (0.000860)			
.× log Weighted Distance _{f,t-1}						0.000242 (0.000761)		
.× Import Intensity _{f,t-1}							0.0304*** (0.00282)	
.× Processing Trade Share _{f,t-1}								-0.0179*** (0.00130)
Obs.	784,347	794,041	514,776	689,754	689,477	2,057,240	724,125	2,150,706
Adj. R ²	0.216	0.216	0.213	0.216	0.216	0.219	0.218	0.221

Notes: Standard errors clustered at the firm–country level are reported in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively. All exercises use specification (21) and include the specified fixed effects.

Table A11: Entry Decisions for Non-Targeted Imports: New Products and Size Measures

	Dependent variable: Entry _{fh<i>i</i>,t}							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
AD _{f,t-1}	-0.0211*** (0.00358)	-0.0173*** (0.00350)	-0.00830*** (0.00204)	-0.0137*** (0.00273)	-0.00124 (0.00114)	0.00890 (0.00741)	-0.00276** (0.00122)	0.00104 (0.000765)
.× log Sales income _{f,t-1}	0.00218*** (0.000370)							
.× log Employment _{f,t-1}		0.00242*** (0.000463)						
.× TFP (OP) _{f,t-1}			0.00867*** (0.00152)					
.× TFP (LP) _{f,t-1}				0.00737*** (0.00131)				
.× TFP (ACF) _{f,t-1}					0.00268*** (0.000972)			
.× log Weighted Distance _{f,t-1}						-0.00127 (0.000876)		
.× Import Intensity _{f,t-1}							0.0136*** (0.00322)	
.× Processing Trade Share _{f,t-1}								-0.00936*** (0.00150)
Obs.	488,040	495,074	326,318	433,636	433,523	1,316,593	454,574	1,363,668
Adj. R ²	0.174	0.173	0.171	0.174	0.174	0.174	0.174	0.176

Notes: Standard errors clustered at the firm–country level are reported in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively. All exercises use specification (21) and include the specified fixed effects.

Table A12: Entry Decisions for Non-Targeted Imports: Existing Products and Size Measures

	Dependent variable: Entry _{fh<i>i</i>,t}							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
AD _{f,t-1}	-0.00816 (0.00987)	-0.000879 (0.00926)	-0.0158*** (0.00605)	-0.0236*** (0.00726)	-0.00681** (0.00288)	0.000529 (0.0152)	-0.0131*** (0.00297)	-0.00368** (0.00181)
.× log Sales income _{f,t-1}	0.000785 (0.00110)							
.× log Employment _{f,t-1}		-0.000281 (0.00127)						
.× TFP (OP) _{f,t-1}			0.0139*** (0.00504)					
.× TFP (LP) _{f,t-1}				0.0116*** (0.00373)				
.× TFP (ACF) _{f,t-1}					0.00708*** (0.00256)			
.× log Weighted Distance _{f,t-1}						-0.00123 (0.00186)		
.× Import Intensity _{f,t-1}							0.0406*** (0.00837)	
.× Processing Trade Share _{f,t-1}								-0.0113*** (0.00302)
Obs.	143,604	145,718	74,243	116,762	116,661	485,554	126,374	522,259
Adj. R ²	0.317	0.315	0.304	0.317	0.317	0.318	0.316	0.319

Notes: Standard errors clustered at the firm–country level are reported in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively. All exercises use specification (21) and include the specified fixed effects.

Table A13: Exit Decisions for Non-Targeted Imports: All Products and Size Measures

	Dependent variable: Exit _{fh<i>i</i>,t}							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
AD _{f,t-1}	-0.00953*** (0.00268)	-0.0101*** (0.00262)	-0.00483*** (0.00153)	-0.00800*** (0.00204)	-0.00338*** (0.000834)	0.0114** (0.00564)	-0.00457*** (0.000872)	-0.00395*** (0.000654)
.× log Sales income _{f,t-1}	0.000765*** (0.000282)							
.× log Employment _{f,t-1}		0.00103*** (0.000346)						
.× TFP (OP) _{f,t-1}			0.00332*** (0.00122)					
.× TFP (LP) _{f,t-1}				0.00281*** (0.00100)				
.× TFP (ACF) _{f,t-1}					0.00108 (0.000701)			
.× log Weighted Distance _{f,t-1}						-0.00165** (0.000675)		
.× Import Intensity _{f,t-1}							0.00633*** (0.00240)	
.× Processing Trade Share _{f,t-1}								0.00404*** (0.00113)
Obs.	784,347	794,041	514,776	689,754	689,477	1,900,539	724,125	1,993,964
Adj. R ²	0.319	0.319	0.333	0.327	0.327	0.280	0.324	0.277

Notes: Standard errors clustered at the firm–country level are reported in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively. All exercises use specification (21) and include the specified fixed effects.

Table A14: Exit Decisions for Non-Targeted Imports: New Products and Size Measures

	Dependent variable: Exit _{fh<i>i</i>,t}							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
AD _{f,t-1}	-0.00806** (0.00334)	-0.00644** (0.00328)	-0.00275 (0.00207)	-0.00524* (0.00270)	-0.00180* (0.00108)	0.0128* (0.00699)	-0.00227** (0.00113)	-0.00259*** (0.000776)
.× log Sales income _{f,t-1}	0.000813** (0.000343)							
.× log Employment _{f,t-1}		0.000786* (0.000427)						
.× TFP (OP) _{f,t-1}			0.00275* (0.00164)					
.× TFP (LP) _{f,t-1}				0.00210 (0.00131)				
.× TFP (ACF) _{f,t-1}					0.000953 (0.000907)			
.× log Weighted Distance _{f,t-1}						-0.00167** (0.000830)		
.× Import Intensity _{f,t-1}							0.00552* (0.00302)	
.× Processing Trade Share _{f,t-1}								0.00365*** (0.00141)
Obs.	488,040	495,074	326,318	433,636	433,523	1,316,593	454,574	1,363,668
Adj. R ²	0.327	0.326	0.342	0.334	0.334	0.276	0.331	0.274

Notes: Standard errors clustered at the firm–country level are reported in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively. All exercises use specification (21) and include the specified fixed effects.

Table A15: Exit Decisions for Non-Targeted Imports: Existing Products and Size Measures

	Dependent variable: Exit _{fh'i,t}							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
AD _{f,t-1}	-0.0187** (0.00765)	-0.0224*** (0.00728)	-0.00525 (0.00402)	-0.0150*** (0.00533)	-0.00280 (0.00214)	0.0212* (0.0124)	-0.00855*** (0.00228)	-0.00617*** (0.00147)
. × log Sales income _{f,t-1}	0.00175** (0.000842)							
. × log Employment _{f,t-1}		0.00266*** (0.000978)						
. × TFP (OP) _{f,t-1}			0.00361 (0.00327)					
. × TFP (LP) _{f,t-1}				0.00682** (0.00269)				
. × TFP (ACF) _{f,t-1}					-0.000460 (0.00185)			
. × Weighted Distance _{f,t-1}						-0.00331** (0.00152)		
. × Import Intensity _{f,t-1}							0.0135** (0.00677)	
. × Processing Trade Share _{f,t-1}								0.00142 (0.00236)
Obs.	143,604	145,718	74,243	116,762	116,661	485,554	126,374	522,259
Adj. R ²	0.311	0.311	0.331	0.328	0.329	0.298	0.318	0.291

Notes: Standard errors clustered at the firm–country level are reported in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively. All exercises use specification (21) and include the specified fixed effects.